Chapter 14

Fueling the Functions: The Digestive System

IN THIS CHAPTER

- **Getting down and dirty with digestion basics**
- **Examining the mouth**
- **Spending time in the stomach**
- **Passing through the intestines and other organs for enzyme digestion**

It's time to feed your hunger for knowledge about how nutrients make their way into the human body. In this chapter, we help you swallow the basics about getting food into the system and digest the details about how nutrients move into blood flow. You also get plenty of practice following the nutritional trail from first bite to final elimination.

Digesting the Basics: It's Alimentary!

REMEMBER Before jumping into a discussion of the alimentary canal, we need to review some basic terms:

- **Ingestion:** Taking in food
- **Mechanical Digestion:** Physically breaking down food into smaller pieces
- **Chemical Digestion:** Changing the composition of food — splitting large molecules into smaller ones — to make it usable by the cells
- **Deglutition:** Swallowing, or moving food from the mouth to the stomach
- **Absorption:** Occurs when digested food moves through the intestinal

wall and into the blood

- **Egestion:** Eliminating waste materials or undigested foods at the lower end of the digestive tract; also known as *defecation*

The digestive tract, or alimentary canal, develops early on in a growing embryo. The primitive gut, or *archenteron*, develops from the *endoderm* (inner germinal layer) during the third week after conception, a stage during which the embryo is known as a *gastrula***.** Whereas the respiratory tract (see Chapter 13) is a two-way street — oxygen flows in and carbon dioxide flows out — the digestive tract is designed to have a one-way flow (although when you're sick or your body detects something bad in the food you've eaten, what goes down sometimes comes back up). Under normal conditions, food moves through your body in the following order:

> Mouth → Pharynx → Esophagus → Stomach → Small intestine → Large intestine

About 8 m long (~26 ft.), the alimentary canal is a muscular tube with a four-layered wall throughout. The outermost layer, the *serosa*, is a thin membrane with cells that secrete a clear, watery fluid to reduce friction as the organs rub against one another. It is also referred to as the *visceral peritoneum*. Proceeding inward, the next layer is the *muscularis*. It contains smooth muscle that contracts to push food through via *peristalsis* (see the section "Stomaching the Body's Fuel" later in this chapter). The outer part of this layer consists of *longitudinal muscle* that runs parallel to the length of the tube. Beneath that is a layer of *circular muscle* that wraps around. The stomach has a third layer of *oblique muscle* to aid in its mixing movements. The next layer is the *submucosa*. The submucosa is mostly areolar, or loose connective tissue to provide space for all of the nerves, glands, and vessels that reside there. The innermost layer is the *mucosa*, creating the continuous lining of the canal. It creates the *lumen*, or space within the tube, and its epithelial tissue is the site of absorption and secretion. In the esophagus, this layer is smooth but in the other organs it folds on itself to varying degrees. The creation of little pits and valleys increases surface area for interaction with the food occupying the lumen.

Chew on these sample questions about the alimentary canal:

1-5 Match the alimentary canal terms with their descriptions.

a. Digestion
b. Ingestion
c. Deglutition
d. Absorption
e. Egestion

1. ______ Taking in food
2. ______ Elimination of waste
3. ______ Movement of food from mouth to stomach
4. ______ Means of transporting food into the blood
5. ______ Mechanical/chemical changing of food composition

6-16 As you read through the rest of this chapter, identify the parts of the digestive system. Referring to Figure 14-1, use the terms that follow to identify the corresponding parts of the digestive system.

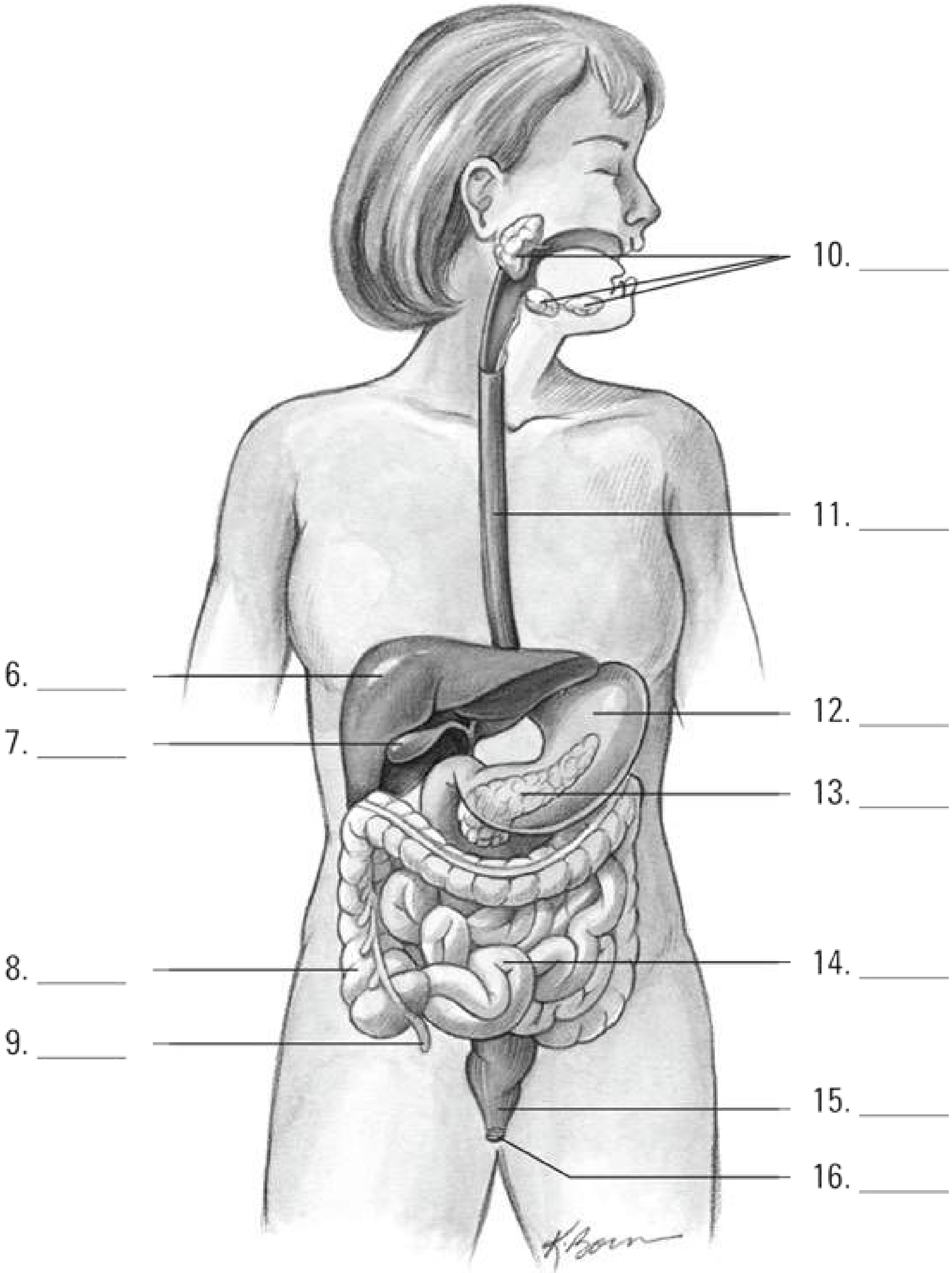


Illustration by Kathryn Born, MA

FIGURE 14-1: The organs and glands of the digestive system.

a. Pancreas
b. Large intestine
c. Liver

d. Small intestine
e. Salivary glands
f. Gallbladder
g. Appendix
h. Anus
i. Esophagus
j. Rectum
k. Stomach

17 Identify the correct sequence of the movement of food through the body:

a. Mouth → Pharynx → Esophagus → Stomach → Small intestine → Large intestine
b. Mouth → Esophagus → Pharynx → Stomach → Small intestine → Large intestine
c. Mouth → Pharynx → Esophagus → Stomach → Large intestine → Small intestine
d. Mouth → Pharynx → Stomach → Esophagus → Small intestine → Large intestine
e. Mouth → Esophagus → Stomach → Small Intestine → Pharynx → Large intestine

18 Indicate the order of the layered walls of the alimentary canal, from the innermost proceeding outward.

______ Muscularis, longitudinal
______ Muscularis, circular
______ Serosa
______ Mucosa
______ Lumen
______ Submucosa

Nothing to Spit At: Into the Mouth and Past the Teeth

In addition to being very useful for communicating, the mouth serves a number of important roles in the digestive process:

- Chewing, formally known as *mastication*, breaks down food mechanically into smaller particles.
- The act of chewing increases blood flow to all the mouth's structures and the lower part of the head.
- Taste buds on the tongue stimulate saliva production. Interestingly, studies have shown that taste preferences can change in reaction to the body's specific needs. In addition, the smell of food can get gastric juices flowing in preparation for digestion.
- Saliva from *salivary glands* in the mouth helps prepare food to be swallowed and begins the chemical breakdown.

The mouth's anatomy begins, of course, with the lips, which are covered by a thin, modified mucous membrane. That membrane is so thin that you can see the red blood in the underlying capillaries. (That's the unromantic reason for the lips' natural rosy glow.) As you see in the following sections, the mouth itself is divided into two regions defined by the *dental arches* of the upper and lower jaws. The *vestibule* is the region between these arches, and the cheeks and lips, whereas the *oral cavity* is the region inside the dental arches. (Figure 14-2 introduces the major structures of the mouth and the pharynx; flip to Chapter 13 for details on the pharynx.)

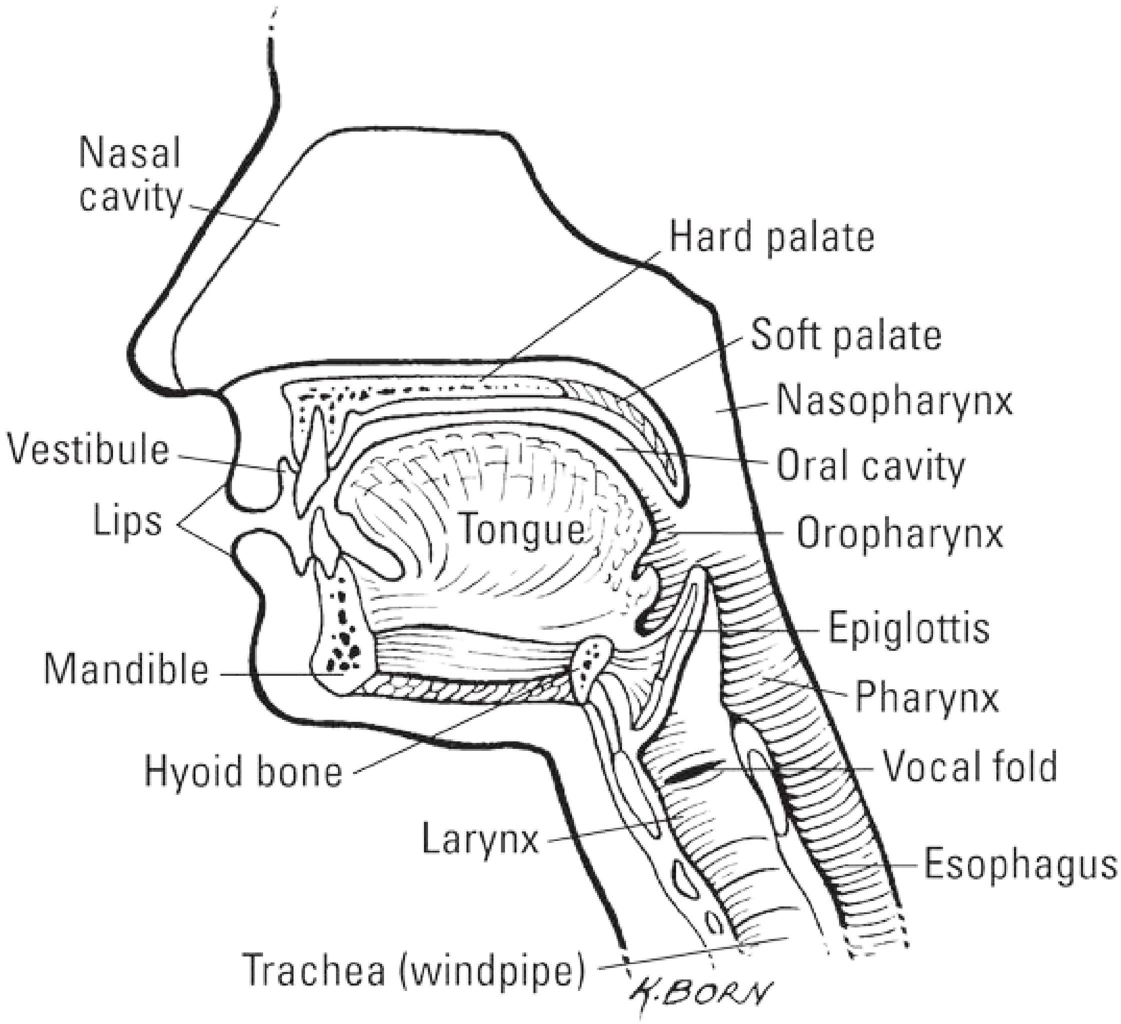


Illustration by Kathryn Born, MA

FIGURE 14-2: Structures of the mouth and pharynx.

Entering the vestibule

The inner surface of the lips is covered by a mucous membrane as well. Sickle-shaped pieces of tissue called *labial frenula* attach the lips to the gums. Within the mucous membrane are *labial glands*, which produce mucus to prevent friction between the lips and the teeth. The cheeks are made up of *buccinator muscles* and a *buccal pad*, a subcutaneous layer of fat, and the *masseters*. The buccinator muscles keep the food between the teeth during the act of chewing while the masseters are the main power behind it. Elastic tissue in the mucous membrane keeps the lining of the cheeks from forming folds that would be bitten during chewing (usually — most people have bitten the insides of their cheeks at one time or another).

The dental arches are formed by the *maxillae* (upper jaw) and the *mandible* (lower jaw) along with the *gingivae* (gums) and teeth of both jaws. The gingivae are dense, fibrous tissues attached to the teeth and the underlying jaw bones; they're covered by a mucous membrane extending from the lips and cheeks to form a collar around the neck of each tooth. The gums are very vascular (meaning that lots of blood vessels run through them) but poorly innervated (meaning that, fortunately, they're not generally very sensitive to pain).

Teeth rise from openings in the jawbone called *sockets*, or *alveoli.* You have a number of different kinds of teeth, and each has a specific contribution to the process of biting and chewing. Most humans get two sets of teeth in a lifetime. The first temporary, or *baby teeth*, set is known as *deciduous teeth.* Babies between 6 months and 2 years old "cut," or *erupt*, four incisors, two canines, and two molars in each jaw. These teeth are slowly replaced by permanent teeth from about 5 or 6 years of age until the final molars — referred to as *wisdom teeth* — erupt between 17 and 25 years of age.

An adult human has the following 16 teeth in each jaw (for a total set of 32 teeth):

- Four *incisors*, which are chisel-shaped teeth at the front of the jaw for biting into and cutting food
- Two *canines*, or *cuspids*, which are pointed teeth on either side of the incisors for grasping and tearing
- Four *premolars*, or *bicuspids*, which are flatter, shallower teeth that come in pairs just behind the canines
- Six *molars*, which are triplets of broad, flat teeth on either side of the jawbone for grinding and mixing food prior to swallowing

REMEMBER Regardless of type, each tooth has three primary parts:

- **Crown:** The part that projects above the gum
- **Neck:** The region where the gum attaches to the tooth
- **Root:** The internal structure that firmly fixes the tooth in the alveolus

(socket)

Teeth primarily consist of yellowish *dentin* with a layer of *enamel* over the crown and a layer of *cementum* over the root and neck, which are connected to the bone by the *periodontal ligament*, or *periodontal membrane*. Cementum and dentin are nearly identical in mineral composition to bone; enamel consists of 94 percent calcium phosphate and calcium carbonate and is thickest over the chewing surface of the tooth.

Depending on the structure of the tooth, the root can be a single-, double-, or even triple-pointed structure. In addition, each tooth has a *pulp cavity* at the center that's filled with connective tissue, nerves, and blood vessels that enter the tooth through the root canal via an opening at the apex of the root of the tooth called the *apical foramen.* Now you know why it hurts so much when dentists have to drill down and take out that part of an infected tooth!

Moving along the oral cavity

The roof of the oral cavity is formed by both the *hard palate,* a bony structure covered by fibrous tissue and the ever-present mucous membrane, and the *soft palate,* a movable partition of fibromuscular tissue that prevents food and liquid from getting into the nasal cavity. (It's also the tissue that sometimes vibrates in sleep, causing a sonorous grating sound referred to as *snoring.*) The soft palate hangs at the back of the oral cavity in two curved folds that form the palatine arches. The *uvula,* a soft conical process (or piece of tissue), hangs in the center between those folds. (You can see the hard and soft palates in Figure 14-2.)

From the soft palate, the oral cavity arches back into the *oropharynx*. Here you'll find a mass of lymphatic tissue on each side — the *palatine tonsils*. That is, if a surgeon hasn't removed them because of frequent childhood infections. The following sections detail the remaining structures found in the oral cavity: the tongue and the salivary glands.

The tongue

As shown in Figure 14-2, the tongue, which is a tight bundle of interlaced muscles, and its associated mucous membrane, forms the floor of the oral cavity. Two distinct groups of muscles — *extrinsic* and *intrinsic* — are used in tandem for mastication (chewing), deglutition (swallowing), and to articulate speech.

- **The extrinsic muscles:** Used to move the tongue in different directions, the extrinsic muscles originate outside the tongue and are attached to the mandible, styloid processes of the temporal bone and the hyoid. A fold of membrane called the *lingual frenulum* anchors the tongue.
- **The intrinsic muscles:** A complex muscle network, the intrinsic muscles allow the tongue to change shape for talking, chewing, and swallowing.

The upper surface of the tongue contains numerous *papillae* (nipple-shaped protrusions) many of which contain taste buds. These contain *gustatory receptors* that relay chemical information to the cortex of the brain.

The salivary glands

The oral cavity has three pairs of salivary glands (which are exocrine glands) whose secretions combine to create *saliva*. Saliva, which contains water, electrolytes, mucus, and enzymes, begins the chemical digestion of the ingested food as well as softens it for ease in swallowing.

- The *parotid* salivary glands are the largest salivary glands and are found in the posterior region of the mandible, just in front of and below each ear. They secrete a serous fluid through the parotid duct (which is opposite the second upper molar tooth) that makes up about 25 percent of the saliva that enters the oral cavity.
- The smallest pair of the trio, the *sublingual* salivary glands, lies near the tongue under the oral cavity's mucous membrane floor. These produce about 5 percent of the saliva that enters the oral cavity.
- The *submandibular* (or *submaxillary*) salivary glands are posterior to the sublingual, along the side of the lower jaw. They're about the size of walnuts and release fluid onto the floor of the mouth, under the tongue. They produce approximately 70 percent of the saliva that enters the oral cavity.

REMEMBER And those secretions are nothing to spit at. Saliva does the following:

- Dissolves and lubricates food to make it easier to swallow

» Contains *amylase*, an enzyme that breaks down certain carbohydrates and *lipase*, which begins lipid (fat) breakdown

» Contains *lysozymes* that break down the cell wall of some bacteria, killing them

» Moistens and lubricates the mouth and lips, keeping them pliable and resilient for speech and chewing

» Frees the mouth and teeth of food, foreign particles, and loose epithelial cells

» Produces the sensation of thirst to prevent you from becoming dehydrated

Following are some practice questions regarding the vestibule and oral cavity:

EXAMPLE **Q.** The function of the mouth is

a. to mix solid foods with saliva.

b. to break down milk protein via the enzyme rennin.

c. To masticate or break down food into small particles.

d. a and c

e. a, b, and c

A. The correct answer is to mix solid foods with saliva and mastication (a and c). The mouth does lots of things, including mixing saliva into the food to add the enzyme amylase, but that's not rennin. With answer options like these, it's best to stick to the basics.

19-28 Use the terms that follow to identify the parts of a tooth, as shown in Figure 14-3.

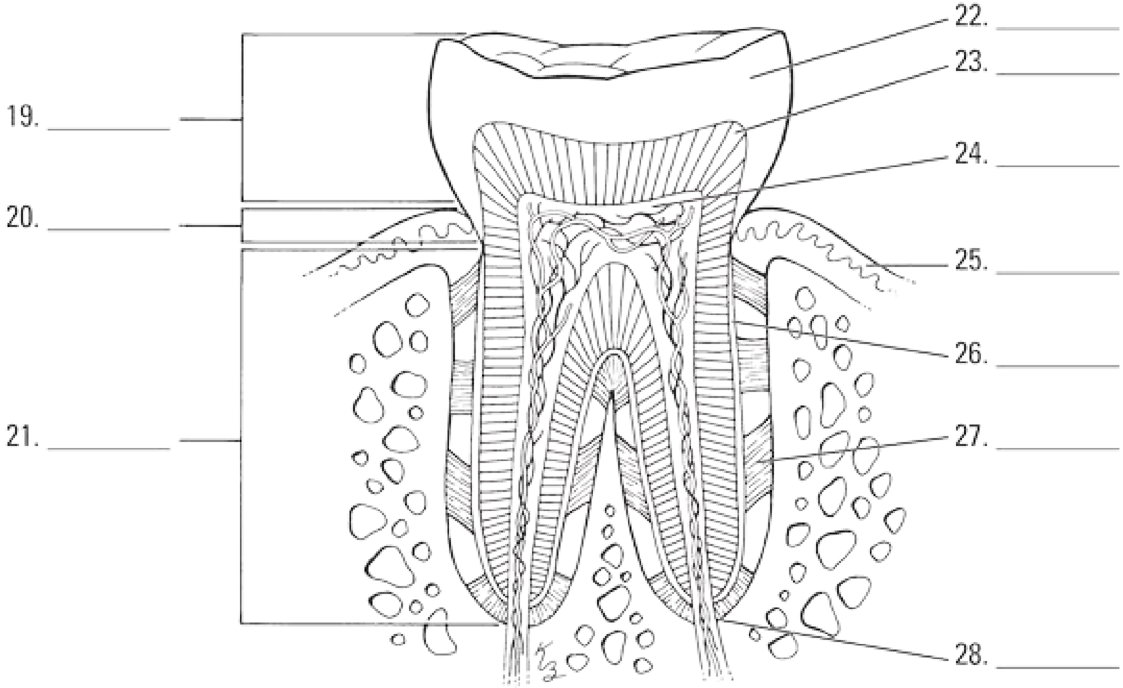


Illustration by Kathryn Born, MA

FIGURE 14-3: The composition of a tooth.

a. Neck
b. Dentin
c. Crown
d. Periodontal ligament
e. Gingiva
f. Enamel
g. Root
h. Apical foramen
i. Pulp cavity
j. Cementum

29 What's the function of the soft palate, which vibrates when you snore?

a. It separates the uvula from the periodontal ligament.
b. It prevents premature swallowing of food before it's thoroughly masticated.

c. It forces saliva back over the tongue.

d. It prevents food and liquid from getting into the nasal cavity.

e. It secretes a watery fluid making the bolus easy to swallow.

30 What anchors the tongue?

a. The hard and soft palates

b. The intrinsic muscles and the sulcus terminalis

c. The extrinsic muscles and the lingual frenulum

d. The gingiva and the labial frenulum

e. The mandibular arches

31 Which of the following statements about teeth is NOT true?

a. The permanent teeth in each human jaw are four incisors, two canines, four premolars, and six molars.

b. Each tooth has a single cuspid anchoring it.

c. They are covered in both dentin and cementum.

d. The tooth cavity contains the tooth pulp.

e. The enamel consists of 94 percent calcium phosphate and calcium carbonate.

32 If it hasn’t been surgically removed, where can the palatine tonsil be found?

a. In the posterior wall of the pharynx

b. In the smooth posterior third of the tongue

c. In the region between the rigid hard palate and the soft palate

d. In the region where the soft palate meets the oropharynx

e. In the region just superior to the uvula

33 Besides making it possible to spit, why do people have saliva?

a. To facilitate swallowing

b. To initiate the digestion of certain carbohydrates

c. To moisten and lubricate the mouth and lips

d. To combat bacteria

e. All of the above

Stomaching the Body's Fuel

REMEMBER *Deglutition*, better known as swallowing, occurs in three phases:

1. **Food pushed into pharynx.** Following mastication, the tongue rolls the food into a *bolus*. It then pushes against hard palate propelling the food into the oropharynx. This is the only step in deglutition that is voluntary. Once the food reaches the oropharynx, the swallowing reflex occurs.
2. **Swallowing reflex.** *Tensor* and *levator* muscles contract causing the uvula and soft palate to raise, blocking off the *nasopharynx*. The hyoid bone and larynx elevate, forcing the *epiglottis* over the *trachea*. This step prevents food from entering the airway but momentarily stops breathing — hence the risk of choking if you gasp or someone makes you laugh. Next, the tongue pushes on the soft palate, sealing off the oral cavity. Now, the only place left for the food to go is into the *esophagus*. The longitudinal muscles of the pharynx contract, pulling the entry into the esophagus up to the bolus. (And you thought swallowing pushed the food down!) The *upper esophageal sphincter* relaxes and the swallowing reflex is complete.
3. **The *bolus* (food mass) heads "down the hatch."** The "hatch" is borrowed nautical slang for the esophagus. The esophagus performs a series of contractions called *peristalsis* to push the food down to the stomach. When the bolus enters the esophagus, it causes the tube to stretch. This triggers the *circular muscles* in the previous ring (the section of esophagus now just above the food) to contract and the next ring (the section now just below the food) to relax. This *receptive relaxation*, along

with the contraction of the *longitudinal muscles* around the food, pushes the food down into the next segment and the process repeats until the bolus reaches the end of the esophagus. The *cardiac (lower esophageal) sphincter* is triggered to relax and the bolus enters the stomach.

After all that pushing and pulling, the food finally (after about 8 seconds, or just 1 if it's liquid) reaches the stomach, a pear-shaped bag of an organ that lies just beneath the ribs and diaphragm. When empty, the *mucosa* of the stomach lies in folds called *rugae,* which allow expansion of the stomach when you gorge and shrink when it's empty to decrease the surface area exposed to acid. Food enters the upper end of the stomach, called the *cardiac* region, through the cardiac sphincter, which generally remains closed to prevent gastric acids from moving up into the esophagus. (When gastric acids do move into places they don't belong, the painful sensation is referred to as "heartburn," which may help you remember the term "cardiac sphincter.") The dome-shaped area next to the cardiac region is called the *fundus;* it expands superiorly (upward) with really big meals. The middle part, or *body,* of the stomach forms a large curve on the left side called the *greater curvature.* The right, much shorter, border of the stomach's body is called the *lesser curvature.* The lower part of the stomach, shaped like the letter J, is the *pylorus.* The far end of the stomach remains closed off by the *pyloric sphincter* until its contents have been digested sufficiently to pass into the *duodenum* of the small intestine.

The secretions of the stomach are released by several glands embedded in the stomach's *epithelium* (the mucosa, or inner lining), which are named for the region in which they are found (cardiac, fundic, and pyloric). There are three main cell types found in these exocrine glands and their secretions combine to form *gastric juice*.

- **Mucous cells:** Secrete mucus to protect the lining from the high acidity of the gastric juices. The main component of mucus is the protein *mucin*.
- **Chief cells:** Secrete *pepsinogen,* a precursor to the enzyme pepsin that helps break down certain proteins into peptides. Chief cells also secrete lipase (to break fats down into lipids) and renin (to break down milk proteins), though production of the latter is often lost by adulthood.
- **Parietal cells** secrete the hydrochloric acid, or HCl, that combines with

pepsinogen to form pepsin to begin protein digestion. HCl also directly breaks down numerous other molecules in the food as well as kills many of the pathogens we inadvertently ingest.

The peristaltic contractions that get the bolus into the stomach aren't limited to the esophagus. Instead, peristalsis continues into the musculature of the stomach and stimulates the release of a hormone called *gastrin*. Within minutes, gastrin triggers secretion of gastric juices that reduce the bolus of food to a thick semiliquid mass called *chyme*, which passes through the pyloric sphincter into the small intestine within one to four hours of the food's consumption. While the main goal of the stomach is mechanical and chemical digestion, a small amount of absorption does occur here. Some water and certain salts find their way into our blood flow through the stomach's mucosa, as do alcohol and lipid-soluble drugs.

REMEMBER Gastric juices are thin, colorless fluids with an extremely acidic pH that ranges from 1 to 4 (see Chapter 2 for details on pH). The quantity of acid released depends on the amount and type of food being digested.

34-41 Use the terms that follow to identify the anatomy of the stomach, as shown in Figure 14-4.

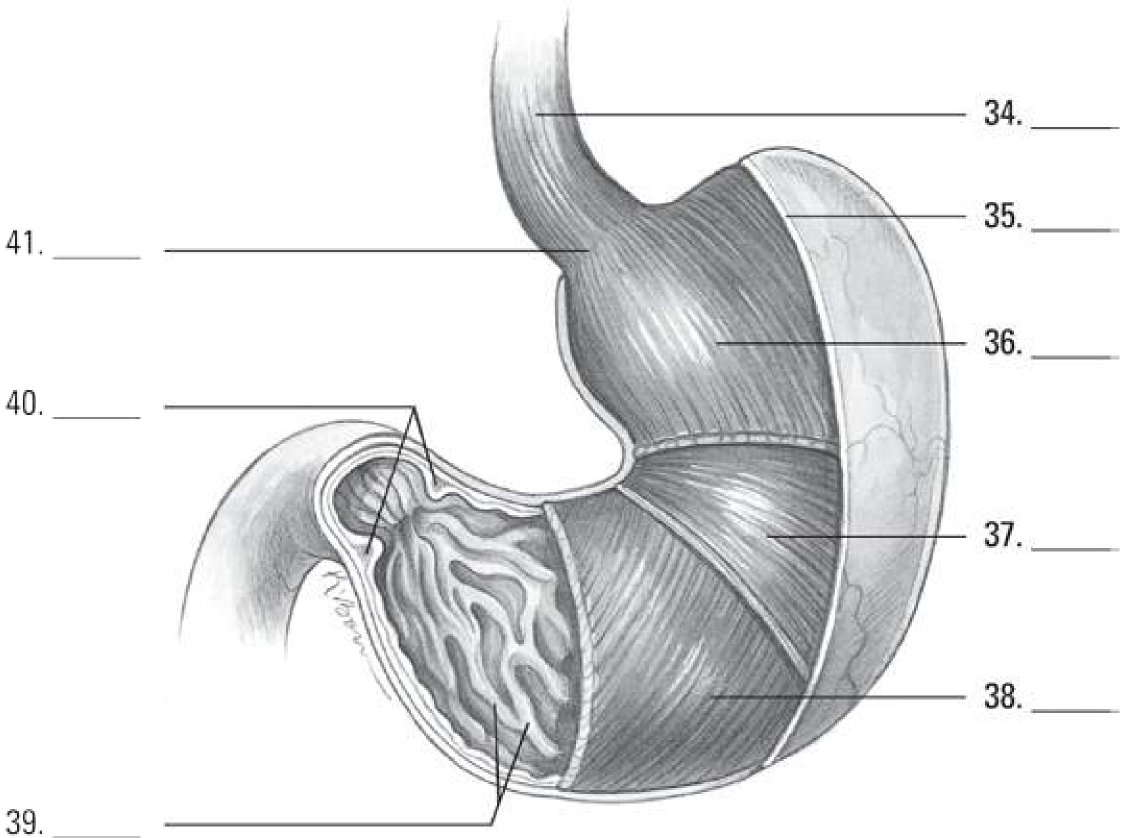


Illustration by Kathryn Born, MA

FIGURE 14-4: The features of the stomach.

a. Circular muscle layer

b. Esophagus

c. Rugae of the mucosa

d. Cardiac (lower esophageal) sphincter

e. Serosa

f. Oblique muscle layer

g. Pyloric sphincter

h. Longitudinal muscle layer

42 What does peristalsis do to food?

a. These contractions of the small intestine wring nutrients from partially digested food.

b. This reflexive upward muscular motion mixes food with stomach acids to create vomit.

c. This grinding action of cartilaginous tissue in the esophagus breaks down food before it enters the stomach.

d. This sequential contraction of circular muscles helps move food through the esophagus.

e. These contractions occur in the colon, hold feces there until defecation.

43 Which part of the digestive tract isn't doing its job when you have heartburn?

a. The esophagus

b. The pyloric sphincter

c. The peritoneal fold

d. The cardiac sphincter

e. The rugae

44 Where can chyme first be found along the digestive tract?

a. The pylorus

b. The peritoneum

c. The jejunum

d. The esophagus

e. The large intestine

Breaking Down the Work of Digestive Enzymes

So what exactly does all the work of digesting and breaking down food? That question brings you back into the realm of proteins (which we introduce in Chapter 2). Proteins that are *enzymes* act as catalysts, meaning that they initiate and accelerate chemical reactions without themselves being permanently changed in the reaction. Enzymes are very picky proteins indeed; they are effective only in their own pH range, they catalyze only a single chemical reaction, they act on a specific substance called a *substrate*, and they function best at 98.6 degrees Fahrenheit, which just happens to be normal body temperature.

The following sections take you on a tour of the organs that produce most of our digestive enzymes.

The small intestine

Most enzymatic reactions — in fact most digestion and practically all absorption of nutrients — takes place in the small intestine. Stretching 7 meters (which is nearly 23 feet!), this long organ extends from the stomach's pylorus to the *ileocecal junction* (the point where the small intestine meets the large intestine), gradually diminishing in diameter along the way.

REMEMBER Three regions of the small intestine play unique roles as chyme moves through them:

- **Duodenum:** The first section of the small intestine is also the shortest and widest section. As partially digested food enters the duodenum, its acidity stimulates the *enterogastric reflex* which feeds back to the stomach. The pyloric sphincter will tighten, preventing the stomach from transporting more chyme than the duodenum can handle. At this time, several hormones are released, though researchers are unsure of the exact role of all of them. Both the liver and pancreas share a common opening into the duodenum. Lined with large and numerous *villi* (fingerlike projections), the duodenum also has *Brunner's glands* that secrete a clear, alkaline mucus. The glands are most numerous near the entry to the stomach and decrease in number toward the opposite, or *jejunum*, end. Nearly all chemical digestion is complete by the time the chyme exits the duodenum.
- **Jejunum:** This is where the bulk of the absorption of nutrients occurs. This region of the small intestine also contains villi, but unlike the duodenum, it has numerous large circular folds at the beginning that decrease in number toward the *ileum* end.
- **Ileum:** Any nutrients not absorbed in the jejunum are absorbed in the

ileum before the food passes into the large intestine. Another feature within the ileum are *Peyer's patches,* which are aggregates of lymph nodes that line this region, becoming largest and most numerous at the distal (far) end. They monitor the gut for pathogenic microorganisms to facilitate the mucosa's generation of immune responses. The ileum opens into the *cecum* of the large intestine through the *ileocecal sphincter.*

A microscopic look at the small intestine reveals circular folds called *plicae circularis,* which project 3 to 10 millimeters into the intestinal lumen, or cavity. These are permanent folds that don't smooth when the intestine is distended. Also present are villi that greatly increase the surface area through which the small intestine can absorb nutrients. Each villus contains a network of capillaries and a central lymph vessel, or *lacteal*, which absorbs fatty acids, to form a milk-white substance called *chyle.* The chyle is carried through lymphatic vessels to the thoracic duct, which empties into the subclavian vein. Simple sugars, amino acids, vitamins, minerals, and water are absorbed by the villi through the mucosa. The surface of the villus is simple columnar epithelium (if you can't recall what that means, flip to Chapter 5) and covered in *microvilli* to further increase surface area. Peristalsis continues into the small intestine, shortening and lengthening the villi to mix intestinal juices with food and increase absorption. Intestinal glands, called the *crypts of Lieberkühn,* lie in the depressions between villi. Packed inside these glands are antimicrobial *Paneth cells, goblet cells* to secrete mucus, and stem cells to facilitate replacement of the epithelial cells.

The extensive curves of the small intestine are held in place by the *mesentery* which also routes the blood and lymphatic vessels as well as nerves. Recent research has shown the mesentery is, in fact, an organ and may play a role in digestive disorders we have not yet pin-pointed a cause for. The *greater omentum* hangs down from the stomach, covering nearly all the intestines. It is laden with fat and contains numerous macrophages to battle pathogens. The greater omentum has been known to migrate to areas of damage, walling the area off to prevent the spread of infection. Both the mesentery and greater omentum are folded extensions of the peritoneum.

REMEMBER Intestinal juices contain three types of enzymes:

- **Enterokinase** has no enzyme action by itself, but when added to pancreatic juices, it combines with *trypsinogen* to form *trypsin*, which can break down proteins.
- **Erepsins, or proteolytic enzymes,** don’t directly digest proteins but instead complete protein digestion started elsewhere. They split polypeptide bonds, separating amino acids; an example is *peptidase*.
- **Inverting enzymes** split disaccharides into monosaccharides as follows:

Enzyme	***Disaccharide***	***Monosaccharides***
Maltase	Maltose	Glucose + Glucose
Lactase	Lactose	Glucose + Galactose
Sucrase	Sucrose	Glucose + Fructose

- **Intestinal lipase** targets fats, splitting them into fatty acids and glycerol.

The liver

The largest internal organ in the body, the liver (shown in Figure 14-5) is divided into a large right lobe and a small left lobe by the *falciform ligament*, another peritoneal fold. Two smaller lobes — the *quadrate* and *caudate* lobes — are found on the lower (inferior) and back (posterior) sides of the right lobe. The quadrate lobe surrounds and cushions the *gallbladder,* a pear-shaped structure that stores and concentrates *bile*, which it empties periodically through the *cystic duct* to the *common bile duct* and on into the duodenum during digestion. This is triggered by the hormone *cholecystokinin (CCK)*, which is released by cells in the duodenum. Bile aids in the digestion and absorption of fats; it consists of bile pigments, bile salts, and cholesterol.

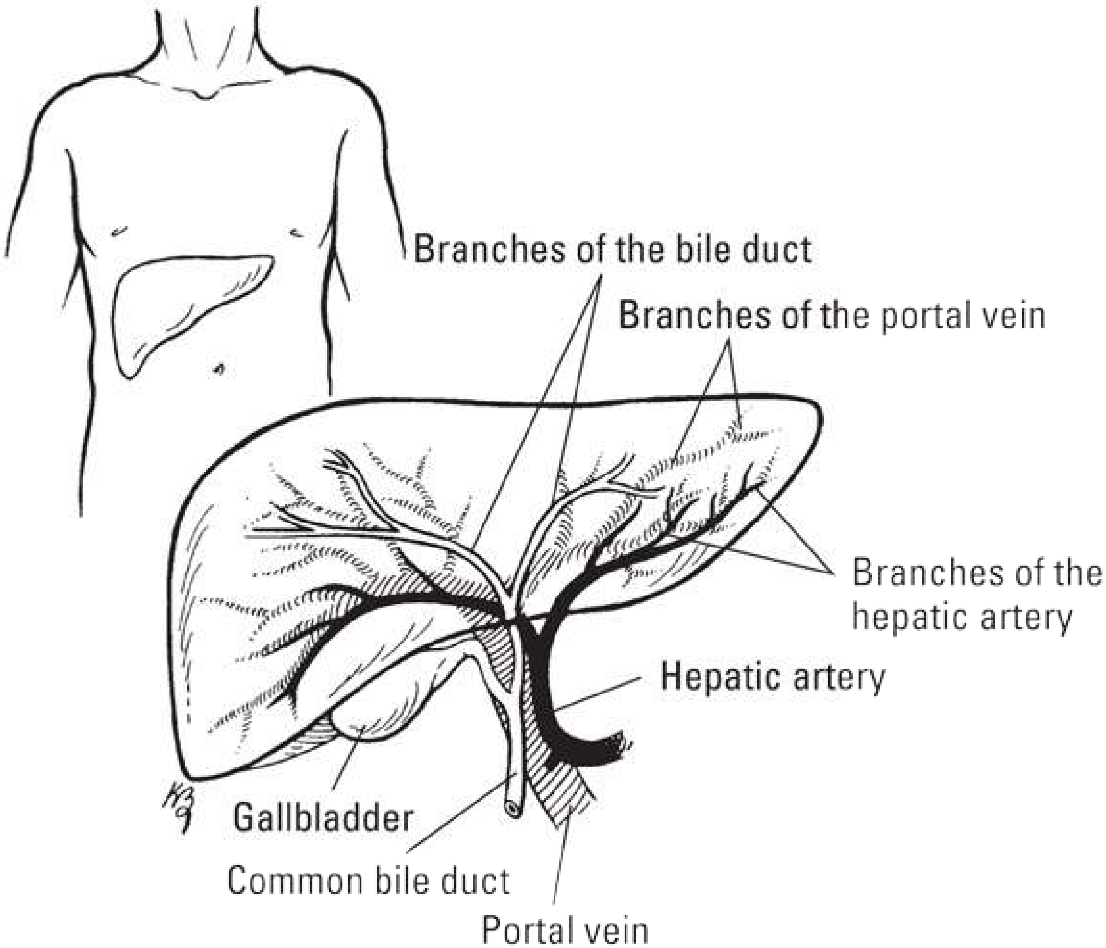


Illustration by Kathryn Born, MA

FIGURE 14-5: A closer look at the liver.

The liver secretes diluted bile through the right and left *hepatic ducts* into the common hepatic duct that joins the cystic duct coming from the gallbladder, forming the common bile duct. The cystic duct allows flow in both directions so bile is usually sent here. Liver tissue is made up of rows of cuboidal cells (see Chapter 5) separated by microscopic blood spaces called *sinusoids* creating the *hepatic lobules,* which are the functional units of the liver. Veins carrying the blood away from the intestines are merged through the mesentery into the *hepatic portal vein*. Thus, all absorbed nutrients are first sent to the liver to be metabolized. Blood from the hepatic artery releases oxygen into the sinusoids so that liver tissue can be oxygenated when processing the absorbed nutrients. The sinusoids also contain *Kupffer cells,* which phagocytize (eat) bacteria.

REMEMBER Considering the number of vital roles the liver plays, its complexity isn't too surprising. Among the liver's various functions are

- Production of blood plasma proteins, including *albumin,* antibodies to fend off disease, a blood anticoagulant called *heparin* that prevents clotting, and bile pigments from red blood cells: the yellow pigment *bilirubin* and the green bile pigment *biliverdin*
- Storage of vitamins and minerals as well as glucose in the form of glycogen
- Conversion and utilization of fats, carbohydrates, and proteins
- Filtering and removal of nonfunctioning red blood cells, toxins, and waste products from amino acid breakdown, such as ammonia

The pancreas

Equally important, though not as large as the liver, the *pancreas* looks like a roughly 7-inch long, irregularly shaped prism. It has a broad head lodged in the curve of the duodenum. The head is attached to the body of the gland by a slight constriction called the neck, and the opposite end gradually tapers to form a tail. The pancreatic duct extends from the head to the tail, receiving the ducts of various lobules that make up the gland. It generally joins the common bile duct, but some 40 percent of humans have a pancreatic duct and a common bile duct that open separately into the duodenum.

REMEMBER Uniquely, the pancreas is both an *exocrine gland,* meaning that it releases its secretion externally either directly or through a duct, and an *endocrine gland,* meaning that it produces hormonal secretions that pass directly into the bloodstream without using a duct. However, most of the pancreas is devoted to being an exocrine gland secreting pancreatic juices into the duodenum. The endocrine portion of the gland secretes insulin, which is vital to the control of sugar metabolism in the body, through small, scattered clumps of cells known as *islets of Langerhans.* Because it contains sodium bicarbonate, pancreatic juice is alkaline, or base, with a pH of 8. The duodenum will trigger the release of alkaline fluid by secreting a hormone called *secretin*. Like the gallbladder and liver, pancreatic enzymes are released in response to CCK. The enzymes released by the pancreas act upon all types of foods, making its secretions the most important to chemical digestion. These include pancreatic amylase, for starch and sugars; pancreatic lipase, for fat breakdown; trypsin and chymotrypsin, for protein breakdown; carboxypeptidase, for peptide breakdown; and nuclease, for nucleic acid breakdown.

That's a lot of material to digest. See how much you remember:

45 Which of the following terms doesn't belong?

a. Enterokinase
b. Maltose
c. Amylase
d. Peptidase
e. Sucrase

46 Which of the following is NOT one of the liver's functions?

a. Production of insulin
b. Production of bile pigments
c. Storage of vitamins and minerals
d. Metabolism of absorbed nutrients
e. Removal of old blood cells

47 Where are most nutrients absorbed?

a. The pylorus
b. The jejunum
c. The ileum
d. The fundus
e. The duodenum

48 What role do villi play throughout the small intestine?

a. They neutralize acids.
b. They propel chyme along the intestinal route.
c. They increase the surface area available to absorb nutrients.
d. They perform mixing movements to increase access to nutrients.
e. They further masticate the chyme.

49 What is the role of enterokinase in digestion?

a. It activates a pancreatic enzyme that can break down protein.
b. It acts as a proteolytic enzyme.
c. It prompts the release of erepsin.
d. It triggers enzyme release from the pancreas and liver.
e. It inverts the functions of other enzymes.

50 What do Paneth and Kupffer cells have in common?

a. They secrete mucus.
b. They secrete digestive enzymes.
c. They propel chyme through their corresponding organs.
d. They both play a role in immunity.
e. They share nothing in common.

51 When the small intestine receives the acidic chyme, the enterogastric reflex occurs causing:

a. the pancreas to release alkaline fluid.
b. the pyloric sphincter to close.
c. the ileocecal sphincter to open.
d. the jejunum to begin peristalsis.
e. the pancreas and gallbladder to release enzymes and bile.

One Last Look Before Leaving

Once chyme enters the large intestine through the *ileocecal sphincter,* it's our last chance to get anything out of it. What's left is packaged into *feces* and excreted from the body via *defecation*.

The appendix

Once considered a *vestigial organ* (one that has no known purpose), the *appendix* is found where the small and large intestine meet. It is true that this tiny, worm-like tube has no digestive function. However, it is comprised of lymphatic tissue and is thought to house our bacteria. This way, we can always repopulate the *intestinal flora* after a bout of diarrhea.

The large intestine

The large intestine, or colon, is about 5 ft. long proceeding up, across, and then down the peritoneal cavity. On its way out of the body, the chyme passes through the following:

Cecum → Ascending colon→ Transverse colon → Descending colon → Sigmoid colon → Rectum → Anus

The large intestine is about 3 inches wide at the start and decreases in width all the way to the rectum. The large intestine has no vili and secretes only mucus so no digestion occurs here. As the unabsorbed material moves through the large intestine, excess water is reabsorbed, drying out the material. In fact, most of the body's water absorption takes place in the large intestine bringing electrolytes along with it.

Bacteria that inhabit the large intestine feed off molecules that we cannot digest like cellulose (plant starch). They use this for energy, creating gas as a byproduct. Though the release of this gas, or *flatus*, can lead to some awkward social situations, the bacteria also synthesize vitamins (K, B12, thiamine, riboflavin) that we absorb through the mucosa. So it's a trade-off really. Perhaps when we pass gas, instead of saying "excuse me," we should say "thank you!"

Peristalsis moves the chyme through the large intestine slowly, segmenting it for maximum water absorption. Following a meal, a *mass movement* occurs where a strong wave of peristalsis pushes the newly formed feces toward the rectum. Once filled, stretch in the walls of the rectum initiates the *defecation reflex*. Thankfully, we can voluntarily inhibit this reflex by contracting the external anal sphincter. When we have prepared ourselves accordingly, we stop contracting the sphincter and allow the reflex to continue. This triggers one final peristaltic wave through the descending colon and feces is forced out of the rectum through the anus.

And that's the end of the line.

52 True or False: The large intestine performs a role in digestion.

53 Which of the following sphincters are under voluntary control?

I. external anal sphincter

II. internal anal sphincter

III. upper esophageal sphincter

a. I
b. III
c. I & III
d. II & III
e. I, II, & III

Answers to Questions on the Digestive Tract

The following are answers to the practice questions presented in this chapter.

1 Taking in food: **b. Ingestion.**

2 Elimination of waste: **e. Egestion.**

3 Movement of food from mouth to stomach: **c. Deglutition.**

4 Means of transporting food into the blood: **d. Absorption.**

5 Mechanical/chemical changing of food composition: **a. Digestion.**

6-16 Following is how Figure 14-1, the digestive system, should be labeled.

6. **c. Liver;** 7. **f. Gallbladder;** 8. **b. Large intestine;** 9. **g. Appendix;** 10. **e. Salivary glands;** 11. **i. Esophagus;** 12. **k. Stomach;** 13. **a. Pancreas;** 14. **d. Small intestine;** 15. **j. Rectum;** 16. **h. Anus**

17 Identify the correct sequence of the movement of food through the body: **a. Mouth → Pharynx → Esophagus → Stomach → Small intestine → Large intestine.**

TIP Although remembering the sequence M-P-E-S-small-large can be helpful, you can also try this phrase to jog your memory: Most Phones Enable Speeches, from Small to Large.

18 Indicate the order of the layered walls of the alimentary canal, from the innermost proceeding outward. **Lumen; Mucosa; Submucosa; Muscularis, circular; Muscularis, longitudinal; Serosa**

WARNING Be careful to note the order requested. A common pitfall is to give them backward and that is often an answer choice.

19-28 Following is how Figure 14-2, the tooth, should be labeled.

19. **c. Crown;** 20. **a. Neck;** 21. **g. Root;** 22. **f. Enamel;** 23. **b. Dentin;** 24. **i. Pulp cavity;** 25. **e. Gingiva;** 26. **j. Cementum;** 27. **d. Periodontal ligament;** 28. **h. Apical foramen**

29 What's the function of the soft palate, which vibrates when you snore? **d. It prevents food and liquid from getting into the nasal cavity.** So, as irritating as snoring may be, you need that soft palate to stay put.

30 What anchors the tongue? **c. The extrinsic muscles and the lingual frenulum.** Extrinsic means "outside," and anchor points occur outside, not

inside (or intrinsically).

31 Which of the following statements about teeth is NOT true? **b. Each tooth has a single cuspid anchoring it.** You can rule out this answer option as false because a cuspid is a type of tooth, so it makes no sense that each tooth would have another type of tooth anchoring it.

32 If it hasn't been surgically removed, where can the palatine tonsil be found? **d. In the region where the soft palate meets the oropharynx.** Superior means above so choice e cannot be correct.

33 Besides making it possible to spit, why do people have saliva? **e. All of the above.** Multifunctional stuff, that saliva. It facilitates swallowing, initiates the digestion of certain carbohydrates, moistens and lubricates the mouth and lips, as well as contains nifty lysosymes to destroy bacteria (the ones that have cell walls anyway).

34-41 Following is how Figure 14-4, the stomach, should be labeled.

34. **b. Esophagus;** 35. **e. Serosa;** 36. **h. Longitudinal muscle layer;** 37. **a. Circular muscle layer;** 38. **f. Oblique muscle layer;** 39. **c. Rugae of the mucosa;** 40. **g. Pyloric sphincter;** 41. **d. Cardiac (lower esophageal) sphincter.**

42 What does peristalsis do to food? **d. This sequential contraction of circular muscles helps move food through the esophagus.**

TIP A bit of Greek may help you remember this term, which comes from the word *peristaltikos,* which means "to wrap around."

43 Which part of the digestive tract isn't doing its job when you have heartburn? **d. The cardiac sphincter.** Biggest clue? Cardiac = heart.

44 Where can chyme first be found along the digestive tract? **a. The pylorus.** Chyme is the thick, semiliquid mass of food that's ready to leave the stomach, so you can select the right answer if you remember your Greek prefixes and suffixes: *pyl*– means "gate," and –*orus* means "guard."

TIP Here's a silly but effective memory tool for this term: When food is ready to leave the stomach, it rings a chime.

45 Which of the following terms doesn't belong? **b. Maltose.** This is a sugar, whereas the other answer options are all enzymes (note the ending –ase).

46 Which of the following is NOT one of the liver's functions? **a. Production of insulin.** That's what the pancreas does.

47 Where are most nutrients absorbed? **b. The jejunum.** The duodenum does digestion (they both start with d) and the ileum is our last-ditch effort to get nutrients out, thus the jejunum does the bulk of the absorption.

TIP To remember this one, keep in mind that each of the three sections of the small intestine plays a unique role. The duodenum neutralizes the stomach acid as the chyme enters the intestine (among other things), the jejunum absorbs most of the nutrients, and the ileum monitors for pathogens (or keeps you from getting "ill" — "il"eum. Get it?).

48 What role do villi play throughout the small intestine? **c. They increase the surface area available to absorb nutrients.** They do other things along the way, but none of those functions are included in the other answers.

49 What is the role of enterokinase in digestion? **a. It activates a pancreatic enzyme that can break down protein.**

TIP It's tricky to remember which of these enzymes is inactive until it combines with something else. You can either try to memorize the function of each enzyme, or you can pick apart the terms. The prefix *entero–* comes from the Greek word for "intestine." The suffix *–kinase* stems from the Greek word for "moving." "Moving through the intestine" sounds like a good guess, don't you think?

50 What do Paneth and Kupffer cells have in common? **d. They both play a role in immunity.**

51 When the small intestine receives the acidic chyme, the enterogastric reflex occurs causing: **b. the pyloric sphincter to close.** While choice a seems reasonable, the release of alkaline fluid is triggered by the hormone secretin and not part of the reflex; the reflex merely controls how much chyme is allowed in at one time.

52 True or false: The large intestine performs a role in digestion. **False.** Every nutrient you're going to get out of the food you eat already has been absorbed by the time it reaches the large intestine. It's the body's primary location for reabsorbing water, but that's not a digestive role. Vitamins you absorb here were created by bacteria, not your digestive processes.

53 Which of the following sphincters are under voluntary control? **a. I.** Thankfully, the sphincter that controls the exit of feces (the external anal sphincter) is under our own control. The upper esophageal sphincter is made of skeletal muscle, leading you to believe that it could be voluntarily controlled, but its relaxation is part of the swallowing reflex and cannot be overridden.

Chapter 15

Filtering Out the Junk: The Urinary System

IN THIS CHAPTER

- **Putting the kidneys on cleanup duty**
- **Following the kidney's role in blood pressure control**
- **Tracking urinary waste out of the body**

If you read Chapter 14 on the digestive system, you may be chewing on the idea that undigested food is the body's primary waste product. But it's not — that title belongs to urine. We make more of it than we do feces — in fact, our bodies are making small amounts of urine all the time — and we release it more often throughout the day. Most important, urine captures all the leftovers from our cells' metabolic activities and jettisons them before they can build up and become toxic. In addition, urine helps maintain the *homeostasis* of body fluids: electrolytes that control acid-base ratio, and the mixture of salt and water that regulates blood pressure.

In short, the urinary system

- Excretes useless and harmful material that it filters from blood plasma, including urea, uric acid, creatinine, and various salts
- Removes excess materials, particularly anything normally present in the blood that builds up to excessive levels
- Maintains proper osmotic pressure, or fluid balance, by eliminating excess water when concentration rises too high at the tissue level
- Reabsorbs water, glucose, ions, and amino acids
- Produces the hormones calcitriol (the hormonally active form of vitamin D that is necessary for calcium absorption) and erythropoietin (which stimulates red blood cell production in the bone marrow), as well as the

hormone renin (that works to maintain blood pressure)

This chapter explains how the urinary system collects, manages, and excretes the waste that the body's cells produce as they go about busily metabolizing all day. You discover the parts of the kidneys, ureter, urinary bladder, and urethra.

Examining the Kidneys, the Body's Filters

The kidneys are nonstop filters that sift through 1.2 liters of blood per minute. Humans have a pair of kidneys just above the waist (lumbar region) toward the back of the abdominal cavity. Although sometimes the same size, the left kidney tends to be a bit larger and more superior than the right due to the size of the liver. The last two pairs of ribs surround and protect each kidney, and a layer of fat, called *perirenal (perinephric) fat*, provides additional cushioning. Kidneys are *retroperitoneal*, which means that they're posterior to the parietal peritoneum. The *renal capsule*, or outer lining of the kidney, is a layer of connective tissue full of collagen fibers; these fibers extend outward to anchor the organ to surrounding structures.

Kidney structure

Each kidney is dark red, about 4½ inches long, and shaped like a bean (hence the type of legumes called kidney beans). The portion of the bean that folds in on itself, referred to as the *medial border*, is concave with a deep depression in it called the *hilum*. The hilum opens into a fat-filled space called the *renal sinus*, which in turn contains the *renal pelvis*, *renal calyces*, blood vessels, and nerves. The *renal artery* and *renal vein*, which provide the kidney's blood supply, as well as the ureter that carries urine to the bladder, leave the kidney through the hilum.

Immediately below the renal capsule is a granular layer called the *renal cortex*, and just below that is an inner layer called the *medulla* that folds into anywhere from 8 to 18 conical projections called the *renal pyramids*. Between the pyramids are *renal columns* that extend from the cortex inward to the renal sinus. The columns allow for passage of the blood vessels. The tips of the pyramids, the *renal papillae*, empty their contents into a collecting area called the *minor calyx.* It’s one of several saclike structures referred to as the *minor* and *major calyces,* which form the start of the urinary tract’s “plumbing” system and collect urine transmitted through the papillae of the pyramids. Although the number varies between individuals, a single minor calyx surrounding the papilla of one pyramid combines into four or five minor calyces, which merge into two or three major calyces. Urine passes through the minor calyx into its major calyx, through the pelvis, and then into the ureter for the trip to the bladder.

Going microscopic

At the microscopic level, each kidney contains more than one million tiny tubes known as *nephrons.* These are the primary functional units of the urinary system. At one end, each nephron is closed off and folded into a small double-cupped structure called a *Bowman’s capsule,* or the *glomerular capsule,* where the actual process of filtration occurs. Leading away from the capsule, the nephron forms into the *proximal convoluted tubule* (PCT), which is lined with cuboidal epithelial cells having microvilli brush borders that increase the area of absorption. This tube straightens to form a structure called the *descending loop of Henle* and then bends back in a hairpin turn into another structure called the *ascending loop of Henle.* After that, the tube becomes convoluted again, forming the *distal convoluted tubule* (DCT), which is made of the same types of cells as the first, or proximal, convoluted tubule but without any microvilli. This tubule connects to a *collecting duct* that it shares with the output ends of many other nephrons. The collecting ducts lie within the pyramids.

Because of their role as the body’s key filters, the kidneys receive about 20 percent of all the blood pumped by the heart each minute. A large branch of the abdominal aorta, called the *renal artery,* carries that blood to the kidneys. After branching into smaller and smaller vessels, the blood eventually moves through interlobular arteries to the *afferent arterioles,* each of which branches into tufts of five to eight capillaries called a *glomerulus* (the plural is *glomeruli*) inside the *glomerular (Bowman’s) capsule*. In the glomerulus, pressure differences force filtration of solutes, fluids, and other glomerular filtrates through the capillary walls into the glomerular (Bowman’s) capsule. The glomerular capillaries come back together to form *efferent arterioles,* which then branch to form the *peritubular capillaries* surrounding the convoluted tubules, the loop of Henle, and the collecting duct. The capillaries come together once again to form a small vein *(venule)* that empties blood into the interlobular vein, eventually moving the blood into the renal vein to depart the kidneys.

Each glomerulus and its surrounding glomerular capsule make up a single *renal corpuscle* where filtration takes place. Like all capillaries, glomeruli have thin, membranous walls, but unlike their capillary cousins elsewhere, these vessels have unusually large pores called *fenestrations* or *fenestrae* (from the Latin word *fenestra* for "window"). Plasma is pushed out here but gas exchange does not take place, which is why the glomerular capillaries merge into an efferent arteriole (as opposed to a vein). Normal capillary exchange takes place in the peritubular capillaries.

Are you getting the flow of all of this? Let's try some questions.

1-6 Use the following terms to identify the parts of the kidney in Figure 15-1.

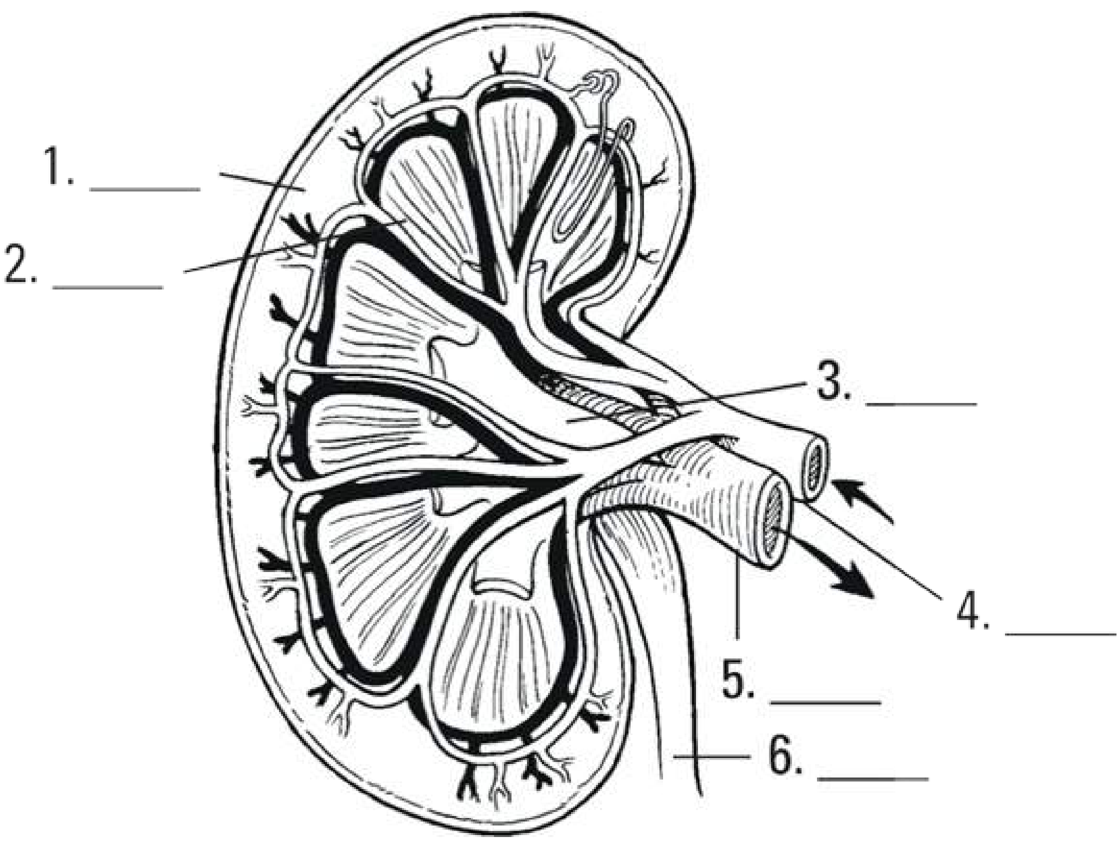


Illustration by Kathryn Born, MA

FIGURE 15-1: Internal anatomy of the kidney.

a. Renal vein

b. Cortex

c. Ureter

d. Renal pelvis

e. Renal artery

f. Medulla (renal pyramids)

7-14 Use the following terms to identify the nephron structures in Figure 15-2.

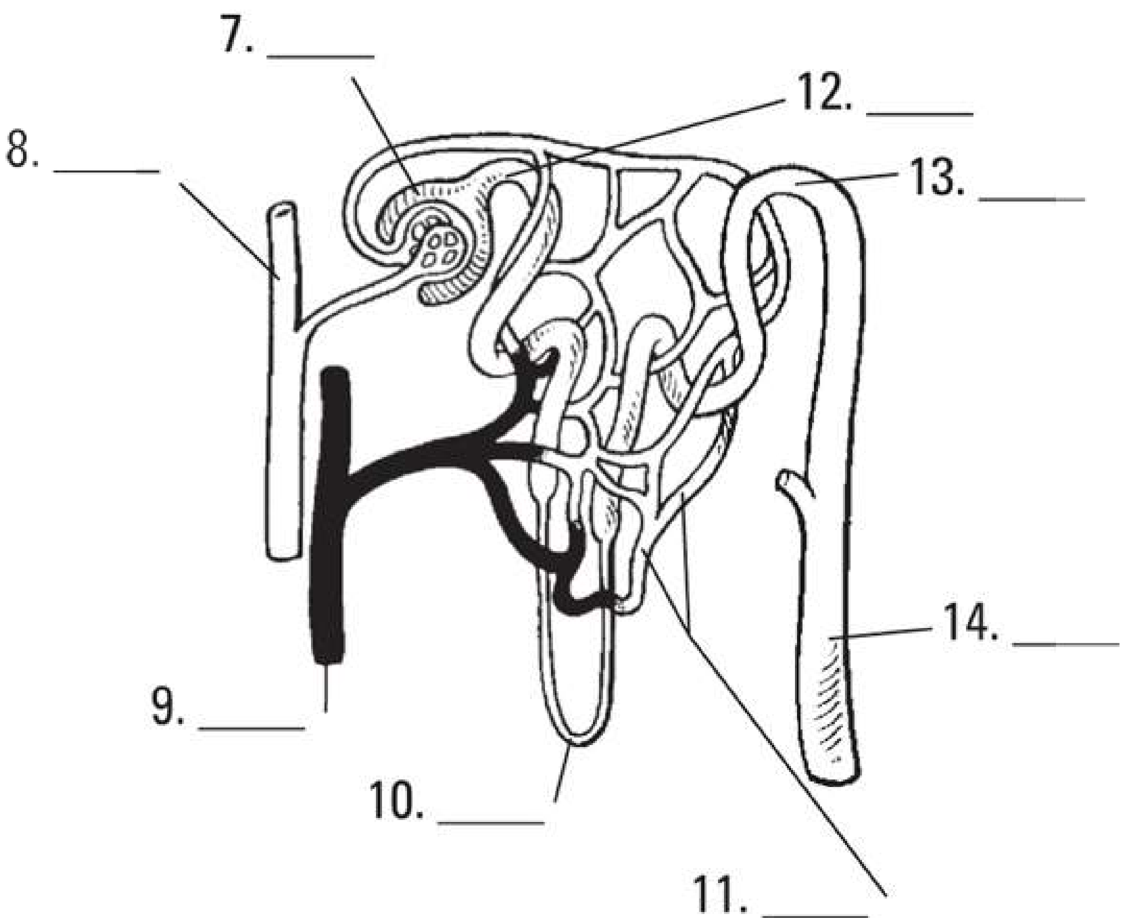


Illustration by Kathryn Born, MA

FIGURE 15-2: Nephron structures.

a. Loop of Henle

b. Glomerular (Bowman's) capsule

c. Collecting duct

d. Proximal convoluted tubule

e. Interlobular vein

f. Peritubular capillary bed

g. Distal convoluted tubule

h. Interlobular artery

15-19 Match the anatomical terms with their descriptions.

a. Composed of folds forming the renal pyramids and the columns in between

b. Granular outer layer

c. Irregular, saclike structures for collecting urine in the renal pelvis

d. Transports urine from the nephrons

e. Funnels urine into the ureter

15. ______ Cortex
16. ______ Medulla
17. ______ Renal pelvis
18. ______ Calyx
19. ______ Collecting duct

20 The human kidney lies outside the abdominal cavity. This makes it

a. retroperitoneal.

b. parietal.

c. visceral.

d. endocoelomic.

e. exterocoelomic.

21 What is the primary functional unit of a kidney?

a. The medulla

b. The loop of Henle

c. The renal papillae
d. The nephron
e. The renal sinus

22 Where are brush borders of villi found primarily?

a. Ascending loop of Henle
b. Proximal convoluted tubule
c. Glomerular (Bowman’s) capsule
d. Descending loop of Henle

23 Although the kidney has many functional parts, where does filtration primarily occur?

a. Inside the distal convoluted tubules
b. Outside the loop of Henle
c. Just beyond the renal cortex
d. In conjunction with the renal sinus
e. Within the glomerulus inside a glomerular (Bowman’s) capsule

Focusing on Filtering

TIP

To understand how the renal corpuscles work, think of an espresso machine: Water is forced under pressure through a sieve containing ground coffee beans, and a filtrate called brewed coffee trickles out the other end. Something similar takes place in the renal corpuscles.

Hydrostatic pressure forces plasma through the holes in the glomerulus, which is referred to as *glomerular filtration*. This filtrate, to the tune of about 125 mL/min., is collected by the surrounding glomerular capsule and ushered into the PCT (see Figure 15-3). So we have a rather effective filter but we can’t afford to lose that much water; we couldn’t possibly drink fast enough to replace it plus we'd urinate constantly. Thus, we have to get most of that water back into the plasma (blood flow). And we do, at 124 mL/min. on average. Furthermore, there are molecules in the filtrate that we'd rather not get rid of such as ions and glucose. That's the job of the nephron — to get back everything in the filtrate that is useful to us.

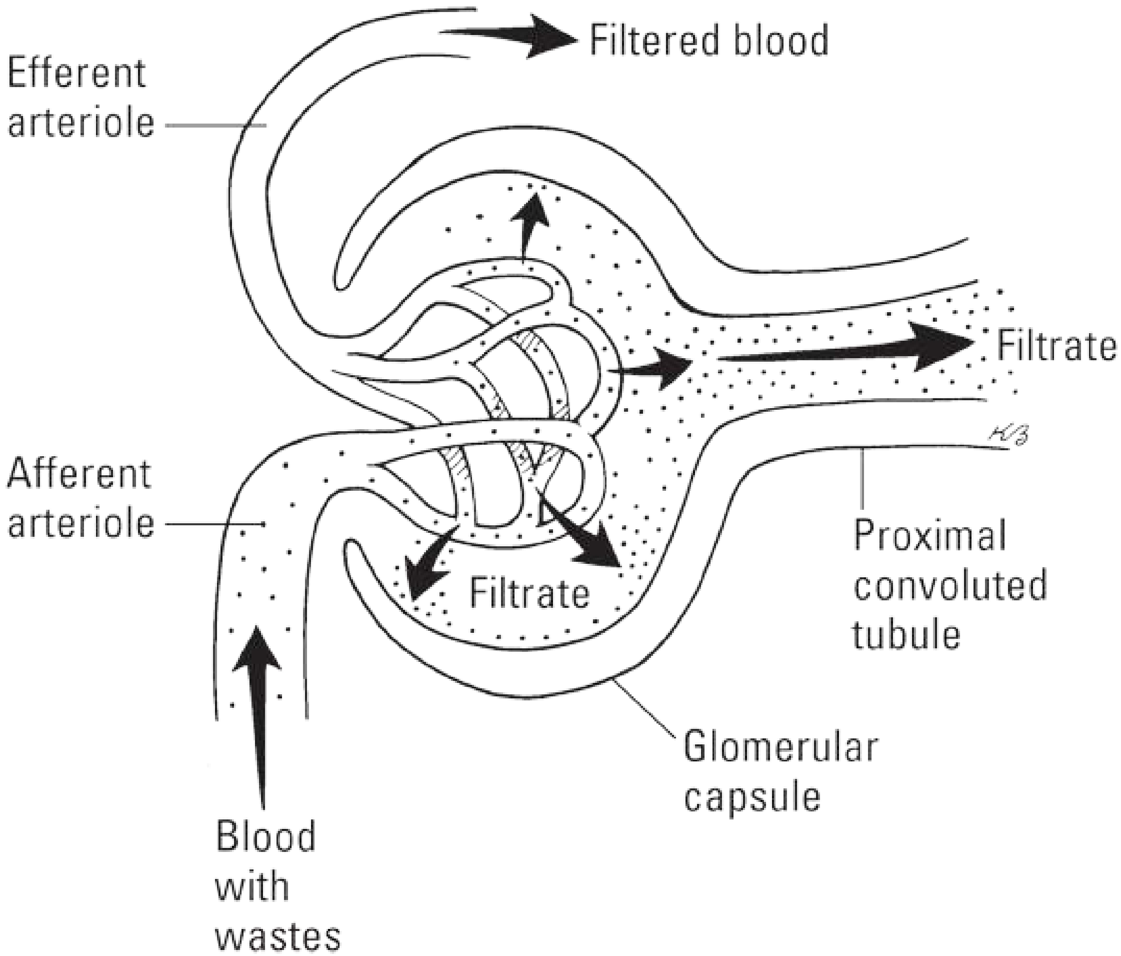


Illustration by Kathryn Born, MA

FIGURE 15-3: Glomerular filtration.

Retaining water

The work of the nephron takes place in three phases, each corresponding to a different section which you can see illustrated in Figure 15-4.

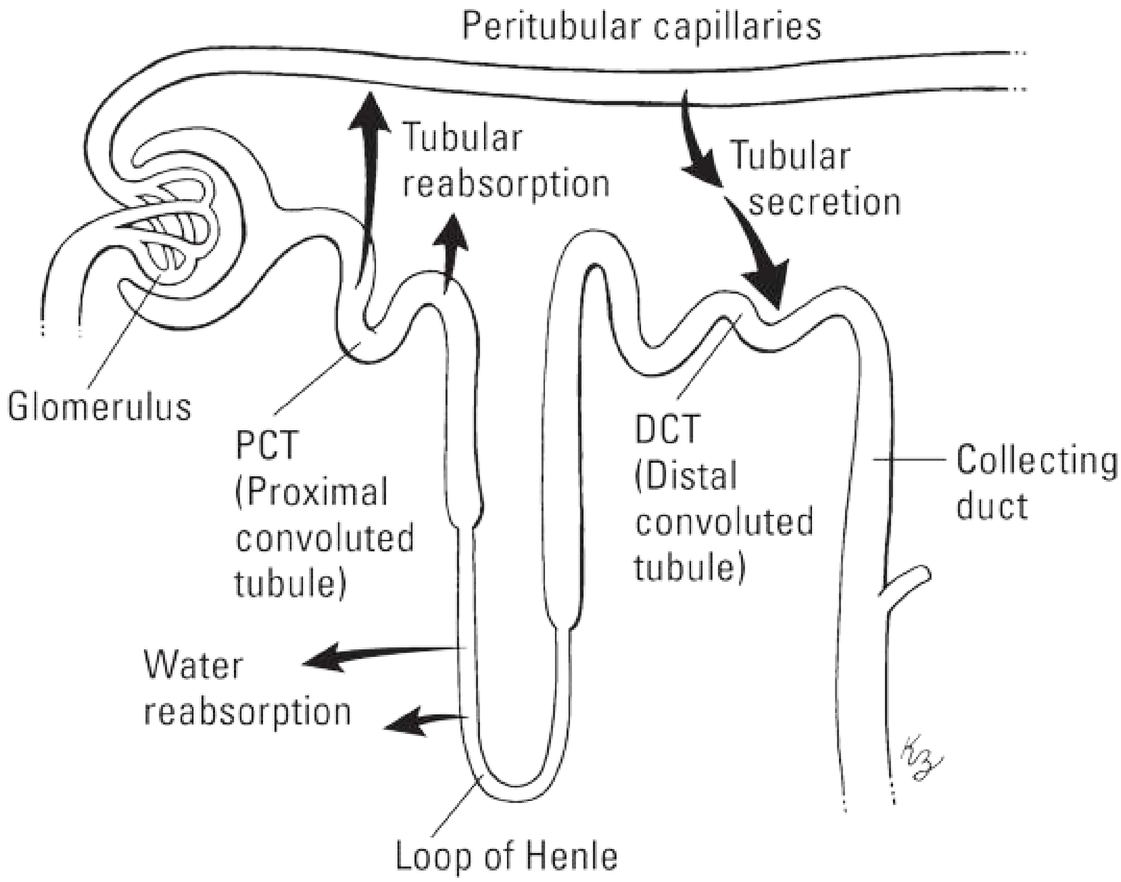


Illustration by Kathryn Born, MA

FIGURE 15-4: Nephron action.

TIP Keep in mind your perspective when reading through the actions of the nephron. In the end, we don't care about what's in the urine, we care about what's in (or isn't in) the blood. So when we say reabsorption, we're talking about reabsorbing back into the blood stream. When we say secretion, we're talking about secreting from the bloodstream.

- **Tubular reabsorption** primarily occurs in the PCT. Molecules that were dissolved in the plasma are now in the filtrate and this is our chance to get them back. Glucose is 100-percent reabsorbed here along with various proteins, amino acids, ions, and of course, water. These molecules are transported out of the filtrate and into the interstitial fluid surrounding the nephron, where they can diffuse into the peritubular capillaries, thus back into blood flow. Wastes such as *urea* and *uric acid* remain in the filtrate.

- **Water reabsorption** primarily occurs in the loop of Henle. Unfortunately, we don't have a mechanism to directly transport water. We can, however, pump ions which can lead to the movement of water by osmosis. Along the ascending loop there are sodium pumps which transport Na^+ out of the filtrate. Anions such as Cl^- follow the sodium ions out, attracted to the opposing charge. In the interstitial fluid, these ions bond forming salts. Consequently, the solute concentration increases — which means the water concentration decreases. Since the concentration of water is now lower outside of the nephron, it moves out. Since so much was pushed out during glomerular filtration, the concentration of water is even lower inside the peritubular capillaries so water flows back into the bloodstream. This is called *countercurrent exchange* and ensures that we are always getting water out of the filtrate by controlling the solute concentration of the interstitial fluid.
- **Tubular secretion** primarily occurs in the DCT. There are two types of wastes that remain in the blood of the peritubular capillaries — those that were too numerous to all be pushed out during filtration (such as hydrogen ions) and those that were too big (such as histamine and drug compounds). Both are targeted by active transport pumps pulling them out of bloodflow and into the filtrate just before it enters the collecting duct.

WARNING Though each nephron action primarily occurs to a specific section, it's important to realize that many of them — especially water reabsorption — can occur anywhere.

Controlling blood volume

While all this filtering and absorption is going on, the kidneys also play a role in maintaining blood pressure by influencing blood volume; more specifically, the amount of water in the blood plasma. When blood pressure drops, not as much filtrate is created so cells in the nephron notice this. They secrete a hormone called *renin*, which kicks off the *renin-angiotensin system* (RAS) — a cascade of reactions that ultimately lead to an increase in blood pressure. To prevent a local change from initiating a global response, the RAS utilizes multiple hormones as a means of checks and balances; renin by itself does nothing. Two other organs must “agree” that blood pressure is too low to initiate the changes that increase it: the liver and the lungs.

The liver, in response to low blood pressure, releases *angiotensinogen* which reacts with renin to create *angiotensin I*, which, by itself, does nothing. The lungs release *ACE (angiotensin converting enzyme),* which coverts angiotensin I into *angiotensin II* and now we can finally address the problem. Angiotensin II causes *vasoconstriction*, or narrowing of the blood vessels. This serves to increase blood pressure but is only a temporary solution. What we really need is to increase the blood volume, which we accomplish by retaining more water in the kidneys. In addition to feeling thirsty (thus prompting you to help the cause by drinking more water), angiotensin II triggers the posterior pituitary to release *ADH (anti-diuretic hormone)* and stimulate the adrenal cortex to release *aldosterone*, both of which head to the kidneys. Aldosterone triggers sodium reabsorption in the DCT, while ADH does so in the collecting duct (which is our last chance to get anything out of the filtrate). Water follows sodium and voilà! More water in the bloodstream, low blood pressure problem solved.

You’ve absorbed a lot in the last few paragraphs. See how much of it is getting caught in your filters.

24 The correct sequence for removal of material from the blood through the nephron is

a. afferent arteriole → glomerulus → proximal convoluted tubule → loop of Henle → distal convoluted tubule → collecting duct.

b. afferent arteriole → glomerulus → distal convoluted tubule → loop of Henle → proximal convoluted tubule → collecting duct.

c. afferent arteriole → collecting duct → glomerulus → proximal convoluted tubule → loop of Henle → distal convoluted tubule.

d. efferent arteriole → proximal convoluted tubule → glomerulus → loop of Henle → distal convoluted tubule → collecting duct.

e. efferent arteriole → glomerulus → proximal convoluted tubule → loop of Henle → distal convoluted tubule → collecting duct.

25 If 125 milliliters of material is captured per minute in the glomerular (Bowman’s) capsules, why is only 1 milliliter of urine generated each minute?

a. The renal pelvis siphons off most of it.
b. A series of tubules allow most of it to be reabsorbed into the bloodstream.
c. The material is highly concentrated by the time it becomes urine.
d. Hydrostatic pressure forces most of the material back out again.

26-29 Match the event to the location in the nephron where it is most likely to occur.

a. renal corpuscle
b. PCT
c. DCT
d. loop of Henle

26. ______ Sodium is actively transported
27. ______ Toxins are secreted
28. ______ Plasma is pushed out
29. ______ Glucose is reabsorbed

30 True or False: Countercurrent exchange occurs when water is actively transported out of the nephron and into the interstitial fluid where it can then passively enter the peritubular capillaries.

31-35 Match the hormone to its origin.

a. posterior pituitary
b. lungs
c. liver
d. kidney
e. adrenal cortex

31. _____ Renin
32. _____ Angiotensinogen
33. _____ Aldosterone
34. _____ ADH
35. _____ ACE

36 Which hormone is responsible for water retention in the kidney from the last place possible before the filtrate becomes urine?

a. renin
b. angiotensin
c. ACE
d. aldosterone
e. ADH

Getting Rid of the Waste

After your kidneys filter out the junk, it's time to deliver it to the bladder and the urethra via the ureters. The following sections explain how that's done.

Surfing the ureters

Ureters are narrow, muscular tubes through which the collected waste travels. About 10 inches long, each ureter descends from a kidney to the posterior lower third of the bladder. Like the kidneys themselves, the ureters are behind the peritoneum outside the abdominal cavity, so the term *retroperitoneal* applies to them, too. The inner wall of the ureter is a simple mucous membrane composed of elastic transitional epithelium. It also has a middle layer of smooth muscle tissue that propels the urine by peristalsis — the same process that moves food through the digestive system. So rather than trickling into the bladder, urine arrives in small spurts as the muscular contractions force it down. The tube is surrounded by an outer layer of fibrous connective tissue that supports it during peristalsis.

Ballooning the bladder

The urinary *bladder* is a large, muscular bag that lies in the pelvis behind the pubis bones. In human females, it's beneath the uterus and in front of the vagina (see Chapter 17). In human males, the bladder lies between the rectum and the symphysis pubis (see Chapter 16). There are three openings in the bladder: two on the back side where the ureters enter and one on the front for the *urethra*, the tube that carries urine outside the body (see the next section). The bladder's *trigone* is the triangular area between these three openings. The *neck* of the bladder surrounds the urethral attachment, and the *internal sphincter* (smooth muscle that provides involuntary control) encircles the junction between the urethra and the bladder. The external urethral sphincter surrounds the urethra nearer to its orifice and this one, thankfully, we have voluntary control over.

Inside, the bladder is lined with highly elastic transitional epithelium (described in Chapter 5). When full, the bladder's lining is smooth and stretched; when empty, the lining lies in a series of folds called *rugae* (just as the stomach does). When the bladder fills, the increased pressure stimulates the organ's stretch receptors, particularly those in the trigone, and initiates the *micturition reflex* (which you feel as the need to urinate). The bladder can hold 600–800 milliliters of urine, but usually the pressure receptor's response will cause it to empty before reaching maximum capacity. (See the later section "Spelling relief: Urination" for more details.)

Distinguishing the male and female urethras

Both males and females have a *urethra*, the tube that carries urine from the bladder to a body opening, or orifice. Both males and females have an internal sphincter controlled by the autonomic nervous system and composed of smooth muscle tissue to guard the exit from the bladder. Both males and females also have an external sphincter composed of skeletal muscle tissue that's under voluntary control. But as we all well know, the exterior plumbing is rather different.

The female urethra is about one and a half inches long and lies close to the vagina's anterior (front) wall. It opens just anterior to (in front of) the vaginal opening. The external sphincter for the female urethra lies just inside the urethra's exit point.

The male urethra is about 8 inches long and carries a different name as it passes through each of three regions:

- The *prostatic urethra* leaving the bladder contains the internal sphincter and passes through the prostate gland. Several openings appear in this region of the urethra, including a small opening where sperm from the vas deferens and ejaculatory duct enters, and prostatic ducts where fluid from the prostate enters.
- The *membranous urethra* is a small 1- or 2-centimeter portion that contains the external sphincter and penetrates the pelvic floor. The tiny *bulbourethral* (or *Cowper's*) *glands* lie on either side of this region.
- The *cavernous urethra*, also known as the *spongy*, or *penile urethra*, runs the length of the penis on its ventral surface through the *corpus spongiosum*, ending at a vertical slit at the end of the penis.

Spelling relief: Urination

Urination, known by the proper term *micturition*, occurs when the bladder is emptied through the urethra. Although urine is created continuously, it's stored in the bladder until the individual finds a convenient time to release it. Mucus produced in the bladder's lining protects its walls from any acidic or alkaline effects of the stored urine. When there is about 200 milliliters of urine distending the bladder walls, stretch receptors transmit impulses to warn that the bladder is filling. *Afferent* impulses are transmitted to the spinal cord, and *efferent* impulses return to the bladder, forming a reflex arc that causes the internal sphincter to relax and the *detrusor muscle* of the bladder to contract, forcing urine into the urethra. The afferent impulses continue up the spinal cord to the brain, creating the urge to urinate. Because the external sphincter is composed of skeletal muscle tissue, no urine usually is released until the individual voluntarily opens the sphincter.

Now test your knowledge of how the human body gets rid of its waste:

EXAMPLE **Q.** The separation of the reproductive and urinary systems is complete in the human

a. male.

b. female.

c. male and female.

A. The correct answer is female. The male urethra runs through the same "plumbing" as the male reproductive system.

37 How can the interior of the bladder expand as much as it does?

a. Its ciliated columnar epithelium can shift away from each other.
b. It is lined with transitional epithelium.
c. Its cuboidal epithelium can realign its structure as needed.
d. Its white fibrous connective tissue gives way under pressure.

38 How does urine move from the kidney down the ureter?

a. By fibrillation
b. By flexure
c. Simply because of gravity
d. By peristaltic contractions
e. It simply flows through.

39 The external sphincter of the urinary tract is made up of

a. circular striated (skeletal) muscle.
b. rugae.
c. simple mucous membrane.
d. the membranous urethra.
e. smooth muscle.

40 What is the function of the internal sphincter at the junction of the bladder neck and the urethra?

a. To stimulate the expulsion of urine from the bladder
b. To keep foreign substances and infections from entering the bladder
c. To prevent urine leakage from the bladder

d. To prevent return of urine from the urethra back into the bladder

e. To relax when voluntarily triggered to expel urine

41 True or False: Urine production occurs sporadically throughout the day.

Answers to Questions on the Urinary System

The following are answers to the practice questions presented in this chapter.

1-6 Following is how Figure 15-1, the internal anatomy of the kidney, should be labeled.

1. **b. Cortex;** 2. **f. Medulla (renal pyramids);** 3. **d. Renal pelvis;** 4. **e. Renal artery;** 5. **a. Renal vein;** 6. **c. Ureter**

7-14 Following is how Figure 15-2, the nephron, should be labeled.

7. **b. Glomerular (Bowman's) capsule;** 8. **h. Interlobular artery;** 9. **e. Interlobular vein;** 10. **a. Loop of Henle;** 11. **f. Peritubular capillary bed;** 12. **d. Proximal convoluted tubule;** 13. **g. Distal convoluted tubule;** 14. **c. Collecting duct**

15 Cortex: **b. Granular outer layer**

16 Medulla: **a. Composed of folds forming the renal pyramids and the columns in between**

17 Renal pelvis: **e. Funnels urine into the ureter**

18 Calyx: **c. Irregular, saclike structures for collecting urine in the renal pelvis**

19 Collecting tubule: **d. Transports urine from the nephrons**

20 The human kidney lies outside the abdominal cavity. This makes it **a. retroperitoneal.** *Peritoneal* refers to the peritoneum, the membrane lining the abdominal cavity; and *retro* can be defined as "situated behind."

21 What is the primary functional unit of a kidney? **d. The nephrons.** Each nephron contains a series of the parts needed to do the kidney's filtering job.

22 Where are brush borders of villi found primarily? **b. Proximal convoluted tubule.** Those brush borders provide extra surface area for reabsorption, so it makes sense that they congregate in the first area after filtration.

23 Although the kidney has many functional parts, where does filtration primarily occur? **e. Within the glomerulus inside a glomerular (Bowman's) capsule.** The glomerulus is a collection of capillaries with big pores, so think of it as the initial filtering sieve.

24 The correct sequence for removal of material from the blood through the nephron is **a. afferent arteriole → glomerulus → proximal convoluted tubule → loop of Henle → distal convoluted tubule → collecting tubule.** In short, blood comes through the artery (arteriole) and material gloms onto the nephron before twisting through the near (proximal) tubes, looping the loop, twisting through the distant (distal) tubes, and collecting itself at the other end. Try remembering artery-glom-proxy-loop-distant-collect.

25 If 125 milliliters of material is captured per minute in the glomerular (Bowman's) capsules, why is only 1 milliliter of urine generated each minute? **b. A series of tubules allow most of it to be reabsorbed into the bloodstream.** Remember the subtraction puzzle we discuss earlier in this chapter? We count up 124 milliliters of reabsorption.

26 Sodium is actively transported: **d. loop of Henle**

27 Toxins are secreted: **c. DCT**

28 Plasma is pushed out: **a. renal corpuscle.** Corpuscle is the term for the glomerulus and the Bowman's capsule together.

29 Glucose is reabsorbed: **b. PCT**

30 Countercurrent exchange occurs when water is actively transported out of the nephron and into the interstitial fluid, where it can then passively enter the peritubular capillaries. **False.** We can't move water, we can only

encourage it to move. To do this, we actively pump Na^+ out of the nephrons, which leads to the formation of salts in the interstitial fluid. Increased solute concentration means decreased water concentration so water exits the nephron passively.

31 Renin: **d. kidney**

32 Angiotensinogen: **c. liver**

33 Aldosterone: **e. adrenal cortex**

34 ADH: **a. posterior pituitary**

35 ACE: **b. lungs**

36 Which hormone is responsible for water retention in the kidney from the last place possible before the filtrate becomes urine? **e. ADH.** The last place possible is the collecting duct which is where ADH triggers ion pumps.

37 How can the interior of the bladder expand as much as it does? **b. It is lined with transitional epithelium.** It's transitional because it needs to be able to stretch and collapse as needed.

38 How does the urine move from the kidney down the ureter? **e. By peristaltic contractions.** It's the same action that moves food through the digestive system.

39 The external sphincter of the urinary tract is made up of **a. circular striated (skeletal) muscle.** The external sphincter is under a person's control. The only tissue in this list of choices that features voluntary control is skeletal muscle.

40 What is the function of the internal sphincter at the junction of the bladder neck and the urethra? **c. To prevent urine leakage from the bladder.** The sphincter's makeup of smooth muscle tissue helps it do so, which means it's not under voluntary control.

41 Urine production occurs sporadically throughout the day. **False.** The kidneys produce urine continuously in a healthy individual.

Part 5

Survival of the Species

IN THIS PART …

Identify the parts of the male reproductive system and explore how chromosomes are packaged to prepare them for their journey.

Follow the female menstrual cycle to understand the nuances of fertility and discover how one cell becomes 32 before the embryo takes up residence in the uterus.

Track the progress from embryo to fetus to newborn baby (technically called a neonate).

Jump from infancy to senescence as the life cycle moves forward.

Chapter 16

Why Ask Y? The Male Reproductive System

IN THIS CHAPTER

- **Explaining the parts of male reproduction**
- **Understanding meiosis and what happens to chromosomes**

Individually, humans don’t need to reproduce to survive. But to survive as a species, a number of individuals must produce and nurture a next generation, carrying their uniqueness forward in the gene pool. Humans are born with the necessary organs to do just that.

In this chapter, you get an overview of the parts and functions of the male reproductive system, along with plenty of practice questions to test your knowledge. (We cover the guys first because their role in the basic reproduction equation isn’t nearly as long or complex as that of their mates. We address the female reproductive system in Chapter 17.)

Identifying the Parts of the Male Reproductive System

On the outside, the male reproductive parts, which you can see in Figure 16-1, are straightforward — a *penis* and a *scrotum.* At birth, the apex of the penis is enclosed in a fold of skin called the *prepuce*, or *foreskin*, which often is removed during a procedure called *circumcision.*

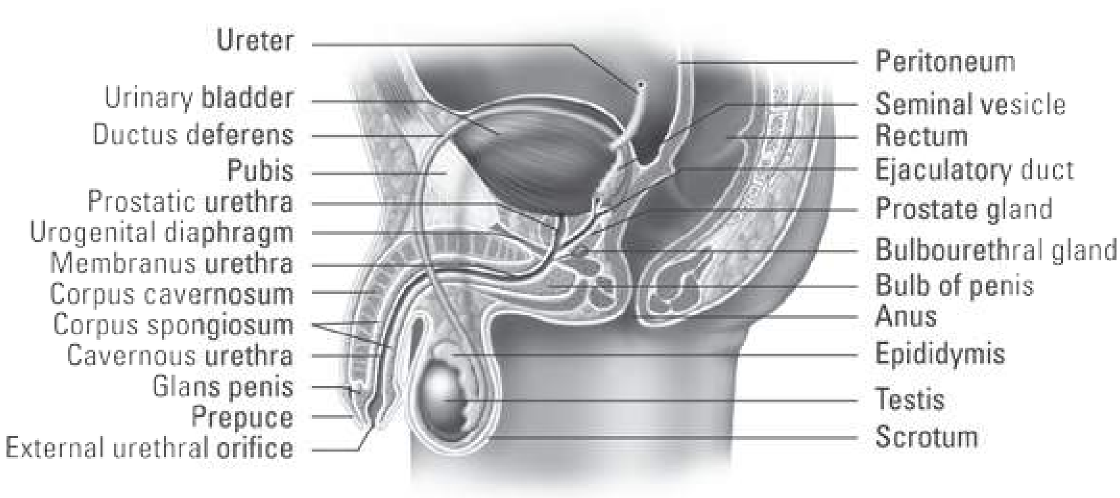


Illustration by Imagineering Media Services Inc.

FIGURE 16-1: The male reproductive system.

The scrotum is a pouch of skin divided in half on the surface by a ridge called a *raphe* that continues up along the underside of the penis and down all the way to the anus. The left side of the scrotum tends to hang lower than the right side to accommodate a longer *spermatic cord,* which we explain later in this section.

There are two scrotal layers: the *integument,* or outer skin layer, and the *dartos tunic,* an inner smooth muscle layer that contracts when cold and elongates when warm. Why? That has to do with the two *testes* (the singular is *testis*) inside (see Figure 16-2). These small ovoid glands, also referred to as *testicles,* need to be a bit cooler than body temperature in order to produce viable sperm for reproduction. When the dartos tunic becomes cold, such as when a man is swimming, it contracts and draws the testes toward the body for warmth. When the dartos tunic becomes overly warm, it elongates to allow the testes to hang farther away from the heat of the body.

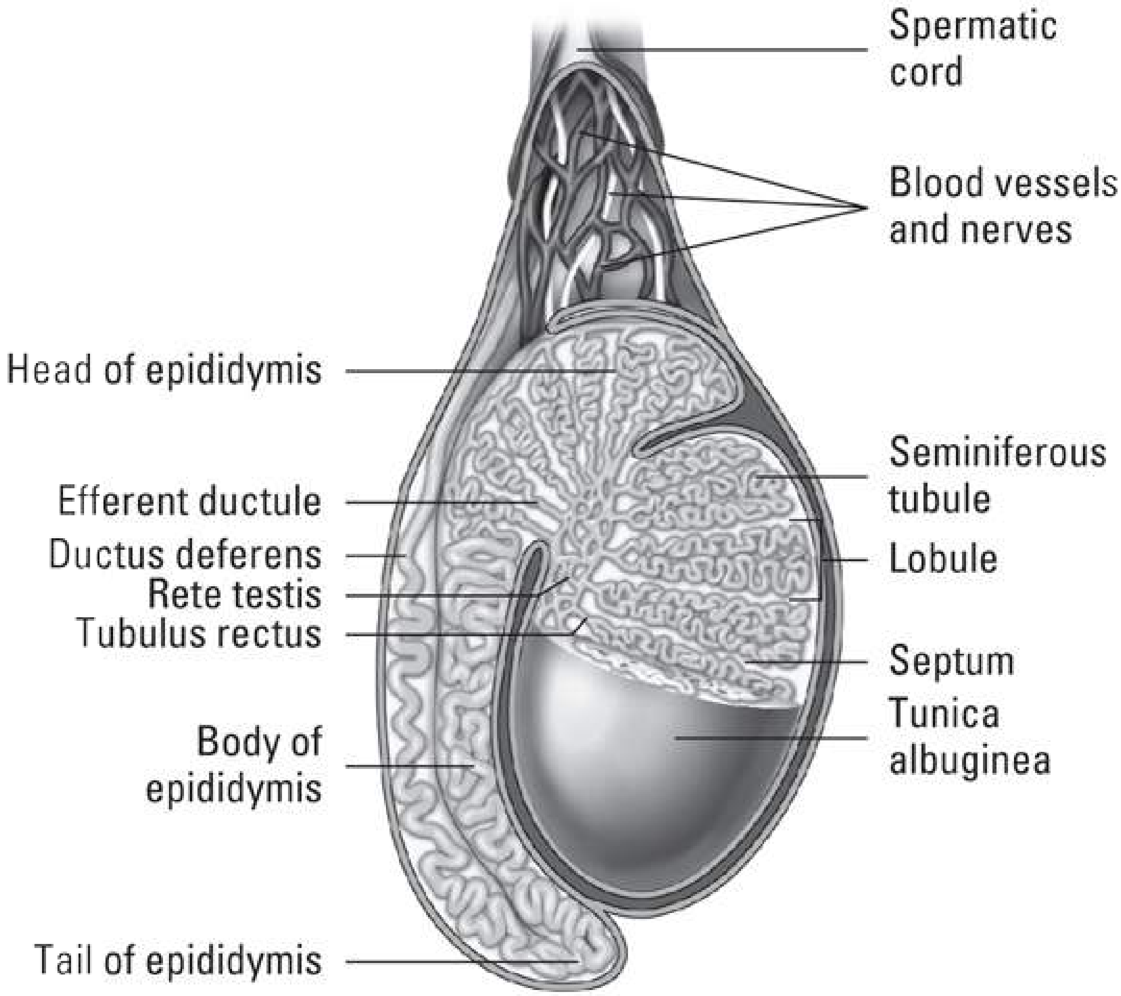


Illustration by Imagineering Media Services Inc.

FIGURE 16-2: Testis.

A fibrous capsule called the *tunica albuginea* encases each testis and extends into the gland forming incomplete *septa* (partitions), which divide the testis into about 200 *lobules.* These compartments contain small, coiled *seminiferous tubules. Spermatogenesis,* or production of sperm cells, takes place in the seminiferous tubules via meiosis. This occurs in the walls of the seminiferous tubules with the newly created gametes, called *spermatids,* beginning their development. By the time they are released into the lumen, now called *spermatozoa,* they are not fully developed, still having a rounded head and a short tail. Spermatogenic (sperm-forming) cells are produced by a series of cellular divisions. These cells, called *spermatogonia,* divide continually by mitosis until puberty. (You can find out more about mitosis in Chapter 4.) Spermatogenesis begins during puberty when a spermatogonium divides mitotically to produce two types of daughter cells:

- **Type A cells** remain in the tubule, producing more spermatogonia.
- **Type B cells** undergo meiosis to produce four sperm, which we review in the next section.

Cells of the seminiferous tubules, called *Sertoli cells* (also known as *sustentacular cells* or *nurse cells*), are activated by follicle stimulating hormone (FSH), which is synthesized and secreted by the anterior pituitary gland**.** (See Chapter 10 for more information on these hormones.) The sustentacular cells secrete androgen-binding protein that concentrates the male sex hormone *testosterone* close to the germ cells to maintain the environment the spermatogonia need to develop and mature**.**

Distributed in gaps between the tubules are *Leydig cells* that produce testosterone. The tubules of each lobule come together in an area called the *mediastinum testis* and straighten into *tubuli recti* before forming a network called the *rete testis* that leads to the *efferent ducts* (also called *ductules*). These ducts carry sperm to an extremely long (about 20 feet), tightly coiled tube called the *epididymis* for storage and maturation of the sperm. This is where sperm fully mature, developing their flagella (though they are not yet able to swim; that does not happen until after ejaculation) and an *acrosome*, the outer casing around the head that contains enzymes to aid in fertilization.

The epididymis merges with the *ductus deferens,* or *vas deferens,* which carries sperm up into the spermatic cord, which also encases the testicular artery and vein, lymphatic vessels, and nerves. Convoluted pouches called *seminal vesicles* lie behind the base of the bladder and secrete an alkaline fluid containing fructose, vitamins, and amino acids to nourish sperm as they enter the *ejaculatory duct* (refer to Figure 16-1). The seminal vesicle also contributes prostaglandins (a hormone) to the semen to trigger muscle contraction once in the female reproductive tract, aiding the sperm's journey toward the ultimate prize.

From there the mixture containing sperm enters the *prostatic urethra* that's surrounded by the *prostate gland.* This gland secretes a thin, opalescent substance that precedes the sperm in an ejaculation, contributing approximately 30 percent of the semen. The alkaline nature of this substance effectively neutralizes the acidity found in the male urethra and reduces the natural acidity of the female's vagina to prepare it to receive the sperm. Two yellowish pea-sized bodies called *bulbourethral glands,* or *Cowper's glands,* lie on either side of the urethra and secrete a clear alkaline lubricant prior to ejaculation; it neutralizes the acidity of the urethra and acts as a lubricant for the penis. Once all the glands have added protective and nourishing fluids to the 400 to 500 million departing sperm, the mixture is known as *seminal fluid* or *semen*.

During sexual arousal, physical and/or mental stimulation causes the brain to send chemical messages via the central nervous system to nerves in the penis telling its blood vessels to dilate so that blood can flow freely into the penis. A *parasympathetic reflex* is triggered that stimulates arterioles to dilate in three columns of spongy erectile tissue in the penis — the *corpus spongiosum* surrounding the urethra and the two *corpora cavernosa,* on the dorsal surface of the erect penis. As the arterioles dilate, blood flow increases while vascular shunts constrict and reroute the blood supply from the bottom to the topside of the penis. Swelling with blood, the penis further compresses its vascular drainage, resulting in a rigid erection that makes the penis capable of entering the female's vagina. Most of the rigidity of the erect penis results from blood pressure in the corpora cavernosa; high blood pressure in the corpus spongiosum could close the urethra, preventing passage of the semen.

Once a sufficient level of stimulation has been received, the *sympathetic nervous system* triggers the two-part process of release. First, rhythmic contractions in the epididymis force sperm through the seminal vesicle and into the prostate gland where they both add their secretions. Then, muscles in the prostate propel the semen into the urethra. This is the process of *emission*. Next, through continuing sympathetic impulse, muscles along the shaft of the penis contract to propel the semen out of the urethra. Once the fluid exits out of the *urethral orifice* (the slit on the *glans penis*, or head of the penis), *ejaculation* has occurred.

See how familiar you are with the male anatomy by tackling these practice questions:

1 Why does the left side of the scrotum tend to hang lower than the right?

a. To adapt to changes occurring during puberty

b. Because the left side contains the larger testicle

c. To accommodate a longer spermatic cord

d. Because most men are right-handed
e. It doesn't, they hang evenly

2 The ______________________ is/are the source of the action when the scrotum adjusts to surrounding temperatures.

a. testes
b. bulbourethral (Cowper's) glands
c. dartos tunic
d. spermatic cord
e. prostatic urethra

3-8 Fill in the blanks in the following sentences.

Sperm-forming cells, or **3.**____________, divide throughout the reproductive lifetime of the male by a process called **4.**_____________. But after puberty, these cells begin to produce two types of **5.**______________ cells. Type A cells remain in the wall of the **6.**______________, producing more of themselves. Type B cells produce **7.**_____________ through the process of **8.**_____________.

9 Where is testosterone produced?

a. Seminiferous tubules
b. Interstitial (Leydig) cells
c. Prostate gland
d. Sertoli (Subtentacular or nurse) cells
e. Epididymis

10 Select the correct sequence for the movement of sperm:

a. Tubuli recti → rete testis → seminiferous tubules → epididymis → ductus deferens → urethra
b. Seminiferous tubules → tubuli recti → rete testis → ejaculatory duct → epididymis → ductus deferens → urethra

c. Epididymis → ejaculatory duct → tubuli recti → rete testis → seminiferous tubules → ductus deferens → urethra

d. Seminiferous tubules → ejaculatory duct → tubuli recti → rete testis → epididymis → ductus deferens → urethra

e. Seminiferous tubules → tubuli recti → rete testis → epididymis → ductus deferens → ejaculatory duct → urethra

11 As the sperm moves through the reproductive tract, which of the following does not add a secretion to it?

a. Leydig (interstitial) cells

b. Cowper's (bulbourethral) glands

c. Seminal vesicle

d. Prostate

12 Sperm storage and development takes place in the convoluted tube called the

a. seminiferous tubule.

b. rete testis.

c. spermatic cord.

d. scrotum.

e. epididymis.

13 After all the proper glands have secreted fluids to nourish and protect the departing sperm, the substance ejaculated is called

a. stroma.

b. semen.

c. prepuce.

d. spermatozoa.

e. inguinal.

14 An average ejaculation will contain about ___________________ sperm.

a. 40 to 50 million
b. 400 to 500 million
c. 400 to 500
d. 400,000 to 500,000
e. 4 to 5 million

15 During sexual arousal, the penis becomes hard and erect. What is happening to cause that?

a. Muscles in the base of the penis are contracting while other muscles toward the end of the penis are expanding.
b. The corpus spongiosum and the corpora cavernosa draw slowly away from one another.
c. Blood vessels are dilating to allow free flow of blood into the penis, while vascular shunts constrict drainage and cause blood pressure to rise in the corpora cavernosa.
d. The Cowper's glands are squeezing additional fluids into the shaft of the penis.
e. A sympathetic impulse triggers the blood vessels to close, trapping blood in the corpus spongiosum.

16 Explain the difference between emission and ejaculation.

____

____

____

____

Packaging the Chromosomes for Delivery

Sperm, the male sex cells, are produced during a process called *meiosis* (which also produces the female sex cell, or *ovum*). Meiosis involves two divisions:

- **Reduction division (meiosis I):** The first, a *reduction division*, divides a single *diploid* cell with two sets of chromosomes into two *haploid* cells with only one set each. These haploid cells have two sets of alleles for each chromosome, though, so another division must occur.
- **Division by mitosis (meiosis II):** The second process is a division by *mitosis* that divides the two haploid cells into four cells with a single set of chromosomes, each with one allele.

Review the reproduction terms in Table 16-1.

Table 16-1 Reproduction Terms to Know

Terms That Vary by Gender	*General Term*	*Male Term*	*Female Term*
Sex organs	Gonads	Testes	Ovaries
Original cell	Gametocyte	Spermatocyte	Oocyte
Meiosis	Gametogenesis	Spermatogenesis	Oogenesis
Sex cell	Gamete	Sperm or Spermatozoa	Ovum

REMEMBER A *diploid cell* (or 2N) has two sets of chromosomes, whereas a *haploid cell* (or 1N) has one set of chromosomes. An allele is the variety, like instructions for attached earlobes versus instructions for free ones.

All cells spend most of their time in *interphase*. While closely associated with mitosis and meiosis (and often mistakenly listed as the first step), this resting phase precedes the division process. In interphase, cells go about their daily business and slowly proceed through steps that prepare them to split in two — namely growth and copying all the DNA. (See Chapter 3 for more on this and other cell terminology). Meiosis, which you can see in Figure 16-3, is described in a series of stages as follows:

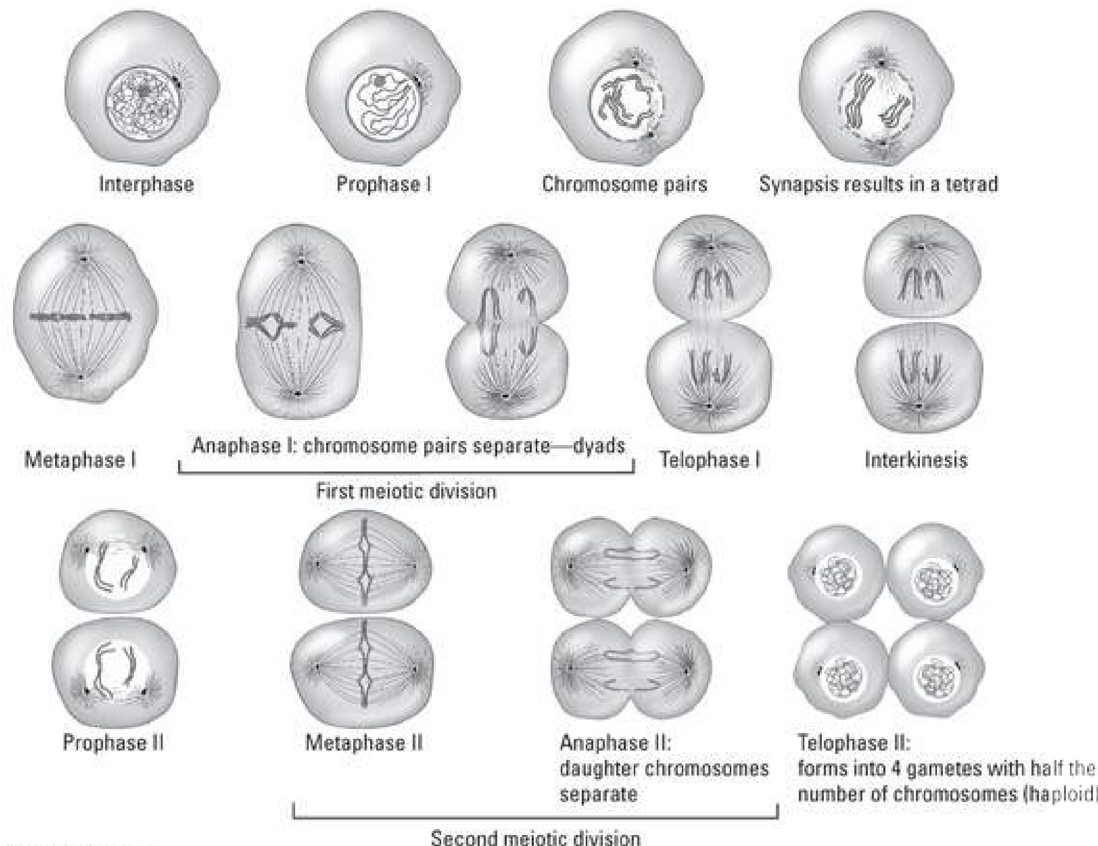
1. **Prophase I:** Structures disappear from the nucleus, including the nuclear

membrane, nucleoplasm, and nucleoli. The cell's centrosomes, each containing two *centrioles*, move to opposite ends of the cell and form poles. Structures begin to appear in the nuclear region, including *spindles* (protein filaments that extend between the poles) and *asters*, or *astral rays* (protein filaments that extend from the poles into the cytoplasm). The *chromatin* (long strands of DNA) condenses to form *chromatids* which then bind together to form the familiar X-shape of the *chromosome*, or *sister chromatids*. *Homologous* chromosomes that contain the same genetic information pair up and go into *synapsis*, twisting around each other to form a *tetrad* of four chromatids. These tetrads begin to migrate toward the *equatorial plane* (an imaginary line between the poles).

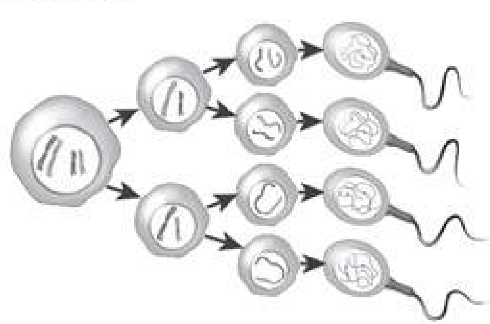
2. **Metaphase I:** The tetrads align on the equatorial plane, attaching to the spindles by the centromere.
3. **Anaphase I:** Homologous chromosomes separate into dyads by moving along the spindles to opposite poles. In late anaphase, a slight furrowing is apparent in the cytoplasm, initiating *cytokinesis* (the division of the cytoplasm). Each pole receives one chromosome (2 identical chromatids).
4. **Telophase I:** The homologous chromosomes are now at opposite poles. Spindle and aster structures disappear, and the nucleus begins to rebuild in each newly forming cell. Chromosomes do not unwind here, they remain as sister chromatids, attached with the *centromere*. The furrowing seen in anaphase I continues to deepen as cytokinesis continues.
5. **Interkinesis:** The cytoplasm fully separates creating two haploid daughter cells. Though each contain one of every chromosome, they are not genetically identical as they can contain different alleles. In the male, the cytoplasm divides equally between the two cells. In the female, cytoplasmic division is unequal creating one cell and one polar body.
6. **Prophase II:** The cells enter the second phase of meiosis. Once again, structures disappear from the nuclei and poles appear at the ends. Spindles and asters appear in the nuclear region. Chromatin is already condensed into chromatids attached by the centromere, and they begin to migrate toward the equatorial plane.
7. **Metaphase II:** The chromatids align on the equatorial plane and the spindles attach to the centromere.

8. **Anaphase II:** The centromere splits and the chromatids separate, moving along the spindles to the poles. A slight furrowing appears in the cytoplasm.
9. **Telophase II:** With the chromatids at the poles, spindles and asters disappear while new nuclear structures appear. The chromatids uncoil, returning to their chromatin form. Cytoplasmic division continues via cytokinesis and each haploid cell divides, forming four cells each with one copy of each DNA strand.

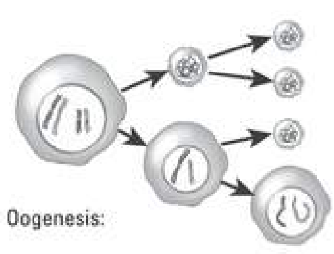
Meiosis:



Spermatogenesis:



In meiosis in male (spermatogenesis) all four haploid cells become functional sperms.



Oogenesis:

In meiosis in female (oogenesis) only one of haploid cells becomes a functional egg.

Illustration by Imagineering Media Services Inc.

FIGURE 16-3: The stages of meiosis.

At the end of this process, the male has four haploid sperm of equal size. As the sperm mature further, a *flagellum* (tail) develops. The female, on the other hand, has produced one large cell, the secondary oocyte, and three small cells called *polar bodies;* all four structures contain just one set of chromosomes. The polar bodies eventually disintegrate and the secondary oocyte becomes the functional cell. When fertilized by the sperm, the resulting *zygote* (fertilized egg) is diploid, containing two sets of chromosomes.

WARNING The terminology in mitosis/meiosis is especially confusing due to multiple uses of the word chromosome. We use the term to refer to our 23 pieces of DNA that contain all our genes. We, however, need two copies of every gene in our cells to function. So we say we have 46 chromosomes total — two copies (with different gene varieties) of each of the 23 pieces. These are not the x-shaped chromosomes we are familiar with though, hence the confusion. Those are created when we copy the 46 pieces of DNA to prepare for division and link them together with their identical match. Our "chromosomes" are really the propeller-shaped chromatids — one complete copy of the piece of DNA. So when we say our cells have two copies of each chromosome, we have two propellers, not two Xs (or four propellers joined in pairs). To further complicate the matter, when cells are not actively preparing for division (and most of the time they aren't), we don't see the propellers either. Chromatids are condensed DNA strands, formed by taking chromatin (the term for the unraveled strand of DNA) and wrapping it around proteins to make the DNA easier to move around. Without this condensation, our cells would be trying to equally split a giant plate of spaghetti which is hard enough as is, but add in the fact that which pieces go to which plate matters, it would be an impossible task without organizing it first.

Think you've conquered this process? Find out by tackling these practice questions:

EXAMPLE **Q.** The metaphase II stage in meiosis involves

a. the slipping of the centromere along the chromosome.

b. the alignment of the chromosomes on the equatorial plane.

c. the contraction of the chromosomes.

d. the disappearance of the nuclear membrane.

A. The correct answer is the alignment of the chromosomes on the equatorial plane. Think “metaphase, middle.”

17 A man has 46 chromosomes in a spermatocyte. How many chromosomes are in each sperm?

a. 23 pairs

b. 23

c. 184

d. 46

e. 46 pairs

18 Synapsis, or side-by-side pairing, of homologous chromosomes

a. occurs in mitosis.

b. completes fertilization.

c. occurs in meiosis.

d. signifies the end of meiosis I.

e. signifies the end of prophase of the second meiotic division.

19 Anaphase I of meiosis is characterized by which of the following?

a. Synapsed chromosomes move away from the poles.

b. DNA duplicates itself and condenses into chromosomes.

c. Synapsis of homologous chromosomes occurs.

d. Homologous chromosomes separate and move poleward with centromeres intact.

e. Chromosomes are split at the centromere and pulled away from the poles.

20 During oogenesis, the three nonfunctional cells produced are called

a. cross-over gametes.

b. spermatozoa.

c. polar bodies.

d. oogonia.

e. somatic cells.

21 Crossing over is a key process in increasing our genetic variability. It occurs when two chromatids (of the same chromosome but with different alleles) basically swap ends, mixing up the varieties of the genes it's carrying. During which phase is this most likely to occur?

a. Prophase II

b. Metaphase II

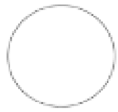
c. Anaphase I

d. Anaphase II

e. Prophase I

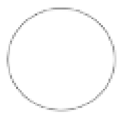
22 Complete the following worksheet on the stages of meiosis. Draw the stages of meiosis and describe the changes in each stage.

1.



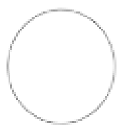
Late Prophase I

2.



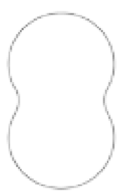
Metaphase I

3.



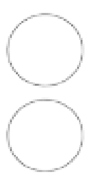
Anaphase I

4.



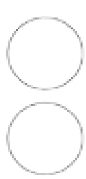
Telophase I

5.

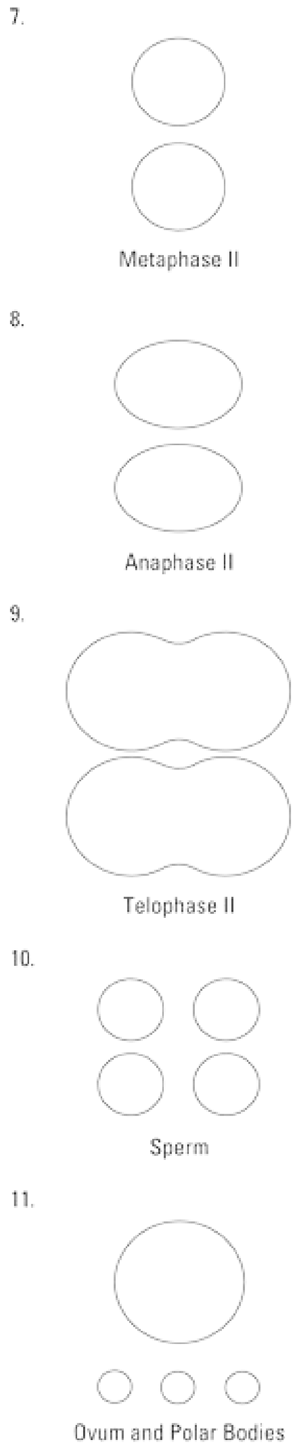


Interkinesis

6.



Prophase II



Answers to Questions on the Male

Reproductive System

The following are answers to the practice questions presented in this chapter.

1 Why does the left side of the scrotum tend to hang lower than the right? **c. To accommodate a longer spermatic cord.** The key words here are "tend to." It's not true of all men, but in general the left testicle hangs lower than the right because it has a longer cord.

2 The **c. dartos tunic** is the source of the action when the scrotum adjusts to surrounding temperatures. This is the inner smooth muscle layer of the scrotum.

3-8 Sperm-forming cells, or **3. spermatogonia,** divide throughout the reproductive lifetime of the male by a process called **4. mitosis.** But after puberty, these cells begin to produce two types of **5. daughter** cells. Type A cells remain in the wall of the **6. seminiferous tubules,** producing more of themselves. Type B cells produce **7. four sperm** through a process called **8. meiosis.**

9 Where is testosterone produced? **b. Leydig (interstitial) cells.** The word *interstitial* can be translated as "placed between," which is where Leydig cells are.

10 Select the correct sequence for the movement of sperm: **e. Seminiferous tubules → tubuli recti → rete testis → epididymis → ductus deferens → ejaculatory duct → urethra.** The sperm develop in the coiled tubules, move through the straighter tubes (tubuli recti), continue across the network of the testis (rete testis), through the efferent ducts, into the epididymis (remember the really long tube), and past the ductus (or vas) deferens and the ejaculatory duct into the urethra.

11 As the sperm moves through the reproductive tract, which of the following does not add a secretion to it? **a. Leydig (interstitial) cells.** Interstitial cells secrete testosterone, the primary male hormone.

12 Sperm storage and development takes place in the convoluted tube called the **e. epididymis.** The other answer options don't come into play until it's time to release semen.

13 After all the proper glands have secreted fluids to nourish and protect the departing sperm, the substance ejaculated is called **b. semen.**

14 An average ejaculation will contain about **b. 400 to 500 million** sperm. Keep in mind that sperm are microscopically small, so quite a few can fit in a tiny amount of semen.

15 During sexual arousal, the penis becomes hard and erect. What is happening to cause that? **c. Blood vessels are dilating to allow free flow of blood into the penis, while vascular shunts constrict drainage and cause blood pressure to rise in the corpora cavernosa.**

16 Explain the difference between emission and ejaculation: **While both are under sympathetic control and seem to happen simultaneously, there is an important distinction. Erection is triggered via a parasympathetic pathway so it's important to note that emission doesn't begin until repeated stimulation initiates it. A simple way to think about it is emission is preparation and ejaculation is release.**

17 A man has 46 chromosomes in a spermatocyte. How many chromosomes are in each sperm? **b. 23.** Because the spermatocyte is the original cell that undergoes division, and because a human has a total of 46 chromosomes only *after* sperm meets egg, you must divide the number 46 in half.

18 Synapsis, or side-by-side pairing, of homologous chromosomes **c. occurs in meiosis.** Specifically, synapsis occurs during prophase I, which is the first stage of meiosis.

19 Anaphase I of meiosis is characterized by which of the following? **d. Homologous chromosomes separate and move poleward with centromeres intact.** Think anaphase, apart. Then you just have to take note of whether it's meiosis I and II to determine what is being pulled apart.

20 During oogenesis, the three nonfunctional cells produced are called **c. polar bodies.** They eventually disintegrate.

21 Crossing over is a key process in increasing our genetic variability. It occurs when two chromatids (of the same chromosome but with different alleles) basically swap ends, mixing up the varieties of the genes it's